

A Preference for Traceable Agency

With a Mutual Information Representation Theorem*

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Abstract

Introduction

To redeem those who lived in the past and to recreate all "it was" into a "thus I willed it" — that alone should I call redemption.

— Friedrich Nietzsche , *Thus Spoke Zarathustra II, "On Redemption"*

The redemption Nietzsche describes is the transformation of a realised outcome ("it was") into something legible as the product of one's choice ("thus I willed it"). We introduce a set of behavioural axioms for ranking choices according to their Nietzschean redemption value and show that they yield the mutual information scale: a family of preferences satisfies the axioms if and only if, in every finite environment, one lottery is weakly preferred to another exactly when it carries at least as much mutual information between act and outcome.

The behaviour captured by our axioms, which we call preference for *traceable agency*, is not merely existential dread. An elected official may prefer a reform whose realised consequences can be traced to the chosen intervention rather than the same distribution of consequences generated by an anonymous budget formula. A donor may prefer a named project to a general fund even when the distribution of social returns is held fixed. In such cases the relevant distinction is not the marginal distribution of final consequences. It is the joint distribution of acts and observations: after the outcome is seen, how much can be learned about which act was realised?

The formal environment is deliberately simple. There is a finite set A of acts, a finite set O of observable outcomes, and a stochastic kernel $P : A \rightarrow \Delta(O)$. A procedure $q \in \Delta(A)$ first draws a realised act A ; conditional on that act, the environment generates O according to P . The primitive behavioural datum is a weak order $\succeq_P$ on $\Delta(A)$ for each fixed environment P . Thus $q \succeq_P q'$ means that, in environment P , procedure q displays at least as much traceable agency as procedure q' .

The main result is an if-and-only-if characterization. A family of preferences $\{\succeq_P\}_P$ satisfies the axioms if and only if, for every finite channel $P : A \rightarrow \Delta(O)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \iff I_{q,P}(A; O) \geq I_{q',P}(A; O),$$

*Email: d.condorelli@gmail.com. Substantial portions of the proof of the main result were developed with assistance from an autonomous, multi-model large-language-model workflow, including generation and cross-checking of mathematical arguments. A Lean 4 verification of Theorem 1 is provided, subject to three known results (Heronst Milnot, Faddeev and Blackwell Theorem), in the appendix and can be replicated using the codebase at [GIT]. All claims, errors, and omissions remain the sole responsibility of the author.

where $I_{q,P}(A; O)$ stands for the mutual information between actions and outcome under channel P at prior q .

The axioms are chosen to express the traceability interpretation directly. The first group is structural. Rankings are weak orders and vary continuously. Duplicating, adding, or deleting irrelevant blocks does not change a comparison. The only strictness requirement in this structural group is local non-triviality: whenever there is genuine uncertainty about which act is realised, full revelation of the act is strictly more traceable than no revelation. This is not a hidden normalization of entropy. It says only that, for every full-support lottery on a non-singleton action set, an outcome that identifies the realised act has strictly more traceability than an outcome that never changes beliefs about it. It fixes the orientation of the criterion and rules out the degenerate one.

An intermediate result shows that this is a family of preferences with DPI.

The next two axioms are monotonicity requirements. If the observed outcome is garbled after it is generated, the act cannot become more traceable. If the act variable itself is replaced by a noisier or coarser description, the resulting procedure cannot display more traceability. The latter axiom is stated for all completions of the induced coarsened channel: rows of the coarsened action variable that are reached with zero probability must not affect the comparison, no matter how they are completed.

The last axiom disciplines composition. Branchwise continuation monotonicity says that, after a first-stage observation, branchwise improvements in continuation traceability should improve the compound procedure, with strictness whenever a positive-probability branch strictly improves. This is a behavioural dominance requirement, not a numerical information equation. Two properties that one might expect to appear as further axioms are instead derived from it, and both derivations turn on the same observation: a first stage that says nothing about the act leaves the continuation comparison untouched. Public-coin independence—replacing both sides of a fixed-prior comparison by the same public mixture with a common background experiment leaves the comparison invariant—is not a separate axiom, because a public coin is such a first stage (Lemma 6). Neither is any separability requirement across statistically independent components: a product channel is a two-stage channel whose first stage is the background experiment, which can teach the observer about the background action while leaving the foreground component independent of it and its marginal unchanged. Branchwise monotonicity therefore already implies that an independent background is behaviourally inert—the ranking of a foreground act–outcome relation coincides with its ranking in isolation, whatever unrelated source of traceability is placed beside it (Lemma 13). This is the only product-environment property the representation argument uses.

The representation has a simple interpretation. Observing O induces a posterior belief about the realised act. The traceability of q in P is the expected reduction in uncertainty about A :

$$I_{q,P}(A; O) = H(q) - \mathbb{E}_{q,P}[H(q(\cdot \mid O))].$$

If outcomes are independent of acts, the value is zero. If the act is known in advance because q is degenerate, the value is also zero. If each act in the support of q produces a distinct observable outcome, the value is $H(q)$. Thus the criterion is not a taste for randomization as such. Randomization matters only because it creates uncertainty about agency for the outcome to resolve.

The theorem characterizes only the traceability component of preference. In applications, under additional additive separability assumptions, it can be combined with an ordinary payoff term, for example

$$V(q, P) = \mathbb{E}_{q,P}[u(O)] + \lambda I_{q,P}(A; O).$$

Axiomatizing the payoff term, the cardinal tradeoff parameter λ , and possible interactions between payoff and traceability would require additional primitives. The present result supplies

the missing behavioural foundation for the information term: once a traceable-agency component is isolated, its scale is forced to be mutual information.

This is made easier by the fact that there are blocks and the scale is universal.

The proof has three steps. First, block coherence, post-processing, and a prior-specific Blackwell argument imply that only the induced law of posterior beliefs about acts matters. Second, the derived public-coin independence property gives affine structure on posterior-law domains, while the derived background inertness of independent components, together with branch and support-face arguments, aligns the initially separate scales. Third, full revelation defines an entropy function; the derived recursion is Faddeev's, and the entropy is therefore Shannon entropy up to the choice of information unit. Mutual information then follows as entropy reduction.

The paper belongs to the decision-theoretic tradition that enriches the object of preference in order to capture motives that expected utility over final payoffs alone leaves out. Kreps [31] and Dekel, Lipman, and Rustichini [16] represent flexibility and subjective uncertainty about future tastes through preferences over menus; Gul and Pesendorfer [26] represent temptation and commitment; Kreps and Porteus [32] enrich the domain to temporal lotteries so that the timing of the resolution of uncertainty can itself carry utility, and Grant, Kajii, and Polak [25] characterise within that domain an intrinsic preference for information, valued independently of any planning use. The freedom-of-choice literature initiated by Pattanaik and Xu [35] ranks opportunity sets by the freedom they embody rather than the consequences they deliver. The motive represented here also has established neighbours on the empirical and applied side. Subjects pay a premium to hold decision rights and to have payoffs depend on their own actions rather than on others' [5, 20, 34]. In the signaling tradition descending from Spence [41], actions are informative about hidden characteristics, and social-image models explain attribution-seeking behaviour, such as named rather than anonymous giving, through observers' beliefs entering utility [24, 8, 2]; self-image models direct the same inference inward, towards beliefs about one's own latent traits [7]; and in Bernheim's conformity model [9] agents instrumentally *suppress* the legibility of their predispositions, an instrumental counterpart of the deniability orientation discussed at the end of the paper. Economics has thus modelled being inferred, being esteemed, and learning about oneself; the trace itself has not been the object of preference.

The present object differs from each neighbour in a specific way. Against the menu and freedom literatures, decision rights and feasible acts are held fixed, and what varies is ex-post authorship rather than ex-ante opportunity: whether the realised outcome statistically records the act. Against Grant, Kajii, and Polak, the information valued flows in the opposite direction—not what signals teach the agent about the world, but what outcomes teach the world about the agent. Against the signaling and image models, the criterion is not instrumental: no audience's beliefs enter utility, and the preference survives anonymity, since what is valued is the trace itself rather than its reputational consequences. Attribution, expressive value, and reputation are possible microfoundations for the behaviour the axioms describe, but they are not primitives of the theorem.

The relation to axiomatic information theory is different in kind. The mathematical characterisation of information measures is a classical subject. Shannon [39] derived entropy from continuity, monotonicity on uniform distributions, and a grouping recursion; Khinchin [29] and Faddeev [19] progressively weakened these postulates, Faddeev reducing them to symmetry, continuity of the binary entropy, and the recursion that appears here as a derived object. Rényi [37] showed that relaxing the grouping postulate to general quasi-arithmetic means yields the one-parameter family bearing his name, locating Shannon entropy as its unique additive member; Aczél and Daróczy [1] give a systematic treatment by functional equations, and Csiszár [14] surveys the field. Mutual information itself is typically characterised indirectly, as entropy reduction or as the relative entropy between the joint law and the product of its marginals: Hobson [28] characterises relative entropy by grouping and additivity conditions, and Baez and

Fritz [4], extending the treatment of entropy as a measure of information loss in Baez, Fritz, and Leinster [6], derive it as the essentially unique functorial, convex-linear, continuous invariant of Bayesian inference.

All of these characterisations take a numerical functional as primitive and impose structural equations on distributions, random variables, or information-loss maps. The present theorem starts instead from ordinal preference relations over lotteries in fixed observation environments and derives the functional: cardinality, product additivity, the chain rule, and finally the Faddeev recursion are consequences of behavioural axioms rather than postulates on a given function. The exercise is thus the information-theoretic analogue of expected-utility theory, in which the numerical form of the criterion is the output of a representation theorem rather than its input; what is characterised is not an abstract measure of information but the preference of an agent for whom that measure is the value of traceable agency.

Blackwell’s comparison of experiments ranks signals by their instrumental value in downstream decision problems [10]; here the ranking is direct, over the trace left by acts in outcomes. Rational-inattention models make information about states a costly input into action [40, 33]. Here information about actions encoded in outcomes is the valued object.

A smaller literature takes information structures themselves as the ranked domain. Gilboa and Lehrer [23] characterise which functions on partitions are values of information for some Bayesian decision problem, and Azrieli and Lehrer [3] extend the characterisation to stochastic structures—the channels ranked here; in both, value is instrumental by construction. Cabrales, Gossner, and Serrano [11] come closest to the present representation from the instrumental side: for ruin-averse investors, the ranking of information structures is represented by entropy reduction, so that mutual information emerges from a specific class of decision problems rather than from ordinal axioms. On the non-instrumental side, Caplin and Leahy [13] put beliefs into the utility function, and Ely, Frankel, and Kamenica [18] derive the entertainment value of suspense and surprise from preferences over the path of beliefs. The present paper combines the two strands: the domain is informational, the valuation is non-instrumental, and, unlike either strand, the functional is not posited or induced by a decision problem but derived ordinally and pinned down uniquely.

The closest formal literature axiomatises the cost of information. Pomatto, Strack, and Tamuz [36] take as primitive a cardinal cost functional on Blackwell experiments, defined without reference to a prior, and show that Blackwell monotonicity, continuity, additivity over independent experiments, and constant marginal cost characterise the log-likelihood ratio costs: weighted sums of Kullback–Leibler divergences between the state-conditional signal distributions. Caplin, Dean, and Leahy [12] and Denti [17] work from state-dependent stochastic choice data generated by rationally inattentive agents and characterise posterior-separable cost functions, with the Shannon cost singled out by invariance conditions; Frankel and Kamenica [21] characterise valid measures of information as expected reductions of measures of uncertainty, with entropy reduction selected by additivity requirements. Throughout this literature, information is instrumental: it concerns an exogenous state and is costly, or valuable, as an input into a downstream decision. This has two structural consequences. Blackwell monotonicity is a theorem or an unquestioned desideratum, since a decision-maker who can condition on a signal is weakly better off with a finer one; and the primitive object is cardinal, or is disciplined by an optimisation model that supplies cardinal structure.

The present paper shares the destination of that literature—coherence over mixing, sequential resolution, and independent combination forces the Shannon scale—but differs in primitive, in direction, and in what is derived rather than assumed. The primitive here is a family of ordinal weak orders over lotteries on the agent’s own acts; there is no downstream decision, and the criterion values what outcomes reveal about the act, not what signals reveal about a state. Prior dependence is essential rather than excluded: the represented functional is the whole map $(q, P) \mapsto I_{q,P}(A; O)$, and the two families are disjoint rather than nested—mutual

information is subadditive over independent experiments and prior-dependent, so it violates the additivity that defines the log-likelihood ratio class, while fully revealing experiments, which have infinite log-likelihood ratio cost, receive the finite value $H(q)$ here. Because the primitive is ordinal and fibred by the prior, the cardinal structure that the cost literature postulates must here be constructed: a public-coin independence property, itself derived from branchwise continuation monotonicity, yields fibrewise affine representations by the mixture-space theorem; the ordinal inertness of independent background components, also derived from branchwise continuation monotonicity, yields only a quasi-additive product form whose interaction term is eliminated by a grouping argument; and the fibrewise scales are aligned into one unit by a cocycle of branch coefficients. Finally, the data-processing conditions (A4) and (A5) are primitive postulates of taste rather than theorems of instrumental rationality; as the sign-duality observation in the Discussion makes precise, their reversal is equally coherent and characterises a preference for deniability. The two literatures therefore meet at the same functional for a structural reason—mutual information is the unique valuation of an information flow coherent in these three respects, whichever end of the channel the valuer occupies—while axiomatising different objects: there, the price of learning about the world; here, a taste for being learned about.

Environment

Let A be a finite set of *actions* and O a finite set of *outcomes*. A channel is a stochastic kernel $P : A \rightarrow \Delta(O)$. For each finite channel P , the primitive behavioural datum is a weak order

$$\succeq_P \quad \text{on} \quad \Delta(A).$$

The interpretation is that $q \succeq_P q'$ means: in the fixed environment P , lottery q displays at least as much traceability as q' . The environment P is not an object of choice. A lottery is the procedure being evaluated; the channel is the fixed observation technology through which the realised act may be recorded by outcomes.

Comparison environments. Let $P : A \rightarrow \Delta(O)$ and $Q : B \rightarrow \Delta(Y)$ be two finite channels. Write A^0 and B^1 for disjoint labelled copies of A and B . The comparison environment

$$P \sqcup Q : A^0 \sqcup B^1 \rightarrow \Delta((O \times \{0\}) \sqcup (Y \times \{1\}))$$

is defined by

$$(P \sqcup Q)((o, 0) \mid a^0) = P(o \mid a), \quad (P \sqcup Q)((y, 1) \mid b^1) = Q(y \mid b),$$

with zero probability assigned to outcomes in the other block. For $q \in \Delta(A)$ and $p \in \Delta(B)$, let $q^0 \in \Delta(A^0 \sqcup B^1)$ and $p^1 \in \Delta(A^0 \sqcup B^1)$ denote the corresponding block-supported lotteries. Comparisons between q under P and p under Q will be written directly as

$$q^0 \succeq_{P \sqcup Q} p^1.$$

This is an ordinary comparison between two lotteries inside the single fixed environment $P \sqcup Q$.

Finite block environments. More generally, let $\{P_i : A_i \rightarrow \Delta(O_i)\}_{i \in K}$ be a finite family of channels, with labelled disjoint copies A_i^i of the action sets. The block environment

$$\bigsqcup_{i \in K} P_i : \bigsqcup_{i \in K} A_i^i \rightarrow \Delta\left(\bigsqcup_{i \in K} (O_i \times \{i\})\right)$$

uses channel P_i on block A_i^i :

$$\left(\bigsqcup_{k \in K} P_k \right) ((o, i) \mid a_i^i) = P_i(o \mid a_i),$$

with zero probability assigned to outcomes in other blocks. For $q_i \in \Delta(A_i)$, write q_i^i for the lottery supported on block A_i^i . Finite block environments are only a notational extension of the two-block construction: they put several otherwise separate act–outcome environments into one fixed comparison problem.

Posteriors and posterior law. Given (q, P) , define the joint law

$$J_{q,P}(a, o) := q(a)P(o \mid a),$$

the outcome marginal

$$m_{q,P}(o) := \sum_{a \in A} q(a)P(o \mid a),$$

and, whenever $m_{q,P}(o) > 0$, the posterior

$$r_o^{q,P}(a) := \frac{q(a)P(o \mid a)}{m_{q,P}(o)}.$$

The posterior law is

$$\mu_{q,P} := \sum_{o: m_{q,P}(o) > 0} m_{q,P}(o) \delta_{r_o^{q,P}}.$$

It is the distribution of beliefs about the realised act induced by the observed outcome; it records both which posterior beliefs can arise and with what probabilities. It is Bayes-plausible:

$$\int_{\Delta(A)} r d\mu_{q,P}(r) = q.$$

Let $\mathsf{X}_A := \Delta_f(\Delta(A))$ be the finite-support posterior-law space on A , and write

$$b(\mu) := \int_{\Delta(A)} r d\mu(r)$$

for the barycentre of $\mu \in \mathsf{X}_A$. For $q \in \Delta(A)$, set

$$\mathcal{M}_q := \{\mu \in \mathsf{X}_A : b(\mu) = q\}.$$

For a nonempty subset $B \subseteq A$, write

$$\Delta_B(A) := \{p \in \Delta(A) : \text{supp}(p) \subseteq B\},$$

and let $\rho_B : \Delta_B(A) \rightarrow \Delta(B)$ be coordinate restriction, $\rho_B(p) = p|_B$. If $\mu \in \mathsf{X}_A$ satisfies $\text{supp}(b(\mu)) \subseteq B$, then $\text{supp}(\mu) \subseteq \Delta_B(A)$: for each $a \notin B$,

$$0 = b(\mu)(a) = \int_{\Delta(A)} p(a) d\mu(p),$$

and non-negativity implies $p(a) = 0$ for every atom p of μ . In this case define the support projection

$$\pi_B(\mu) := (\rho_B)_\# \mu = \sum_{p \in \text{supp}(\mu)} \mu(\{p\}) \delta_{p|_B} \in \mathsf{X}_B.$$

It preserves barycentres on the face:

$$b(\pi_B(\mu)) = b(\mu)|_B.$$

In particular, if $b(\mu) = r$ and $\text{supp}(r) = B$, then $\pi_B(\mu) \in \mathcal{M}_{r|_B}$. When the barycentre assigns zero probability outside B , all posterior beliefs lie on the face $\Delta_B(A)$; π_B identifies those beliefs with distributions on B .

Public-coin mixing. For channels $P : A \rightarrow \Delta(O_P)$ and $Q : A \rightarrow \Delta(O_Q)$ and $\lambda \in [0, 1]$, define

$$\lambda P \oplus (1 - \lambda)Q : A \rightarrow \Delta((O_P \times \{0\}) \sqcup (O_Q \times \{1\}))$$

by

$$(\lambda P \oplus (1 - \lambda)Q)((o, 0) \mid a) = \lambda P(o \mid a),$$

and

$$(\lambda P \oplus (1 - \lambda)Q)((o, 1) \mid a) = (1 - \lambda)Q(o \mid a).$$

Then

$$\mu_{q, \lambda P \oplus (1 - \lambda)Q} = \lambda \mu_{q, P} + (1 - \lambda) \mu_{q, Q}.$$

Because the coin realisation is observed, the observer knows which experiment was used, and the induced posterior law is the corresponding convex mixture of posterior laws.

Outcome post-processing. For a stochastic kernel $T : O \rightarrow \Delta(O')$, define

$$(PT)(o' \mid a) := \sum_{o \in O} P(o \mid a) T(o' \mid o).$$

The kernel T is a garbling applied after the original outcome is generated. It changes the recorded observation while leaving the realised act and the pre-garbling channel P fixed.

Action pushforward. For a stochastic kernel $S : A \rightarrow \Delta(A')$, define

$$(qS)(a') := \sum_{a \in A} q(a) S(a' \mid a), \quad (rS)(a') := \sum_{a \in A} r(a) S(a' \mid a).$$

For $\mu \in \mathbb{X}_A$, write

$$\mu S := \mu \circ (r \mapsto rS)^{-1} \in \mathbb{X}_{A'}.$$

The kernel S reports the realised act through a possibly noisy description. Given $P : A \rightarrow \Delta(O)$ and $q \in \Delta(A)$, the Bayesian pushforward channel is the prior-dependent channel from reported acts to outcomes induced by the joint law $q(a)S(a' \mid a)P(o \mid a)$. Formally, define

$$S^q P : A' \rightarrow \Delta(O)$$

by

$$S^q P(o \mid a') := \frac{\sum_{a \in A} q(a) S(a' \mid a) P(o \mid a)}{(qS)(a')}$$

whenever $(qS)(a') > 0$. On rows a' with $(qS)(a') = 0$, Bayes' rule imposes no value. A completion of $S^q P$ is any channel $\hat{P} : A' \rightarrow \Delta(O)$ that agrees with the displayed formula on every row with $(qS)(a') > 0$. For every such completion,

$$\mu_{qS, \hat{P}} = \mu_{q, P} S,$$

since zero-mass rows receive no probability under qS . When a statement uses $S^q P$ without naming a completion, it is interpreted for every completion of the Bayesian pushforward.

Two-stage composition. Let $P_1 : A \rightarrow \Delta(O_1)$ be a first-stage channel. For each $o \in O_1$, let $Q^o : A \rightarrow \Delta(O_2^o)$ be a continuation channel. Define

$$P_1 \triangleright \{Q^o\}_{o \in O_1} : A \rightarrow \Delta \left(\bigsqcup_{o \in O_1} (\{o\} \times O_2^o) \right)$$

by

$$(P_1 \triangleright \{Q^o\})(o, z \mid a) = P_1(o \mid a) Q^o(z \mid a).$$

The first signal selects a posterior branch; the continuation channel Q^o is the observation technology used within that branch. If q is the initial lottery, write $m(o) := m_{q, P_1}(o)$ and, for each o with $m(o) > 0$, $r_o := r_o^{q, P_1}$. Zero-probability branches ($m(o) = 0$) do not affect the induced posterior law and are ignored throughout.

Product environments. For $P_i : A_i \rightarrow \Delta(O_i)$, define

$$(P_1 \otimes P_2)(o_1, o_2 \mid a_1, a_2) := P_1(o_1 \mid a_1) P_2(o_2 \mid a_2).$$

Product environments combine statistically independent act–outcome components. For $q_i \in \Delta(A_i)$, write $q_1 \otimes q_2$ for the product lottery. For posterior laws

$$\mu = \sum_i m_i \delta_{r_i} \in \mathbb{X}_{A_1}, \quad \rho = \sum_j n_j \delta_{s_j} \in \mathbb{X}_{A_2},$$

write

$$\mu \otimes \rho := \sum_{i,j} m_i n_j \delta_{r_i \otimes s_j} \in \mathbb{X}_{A_1 \times A_2}.$$

Bayes' rule applied to a product prior and a product channel factorises: the joint outcome (o_1, o_2) has probability $m_{q_1, P_1}(o_1) m_{q_2, P_2}(o_2)$ and induces the posterior $q_1^{o_1} \otimes q_2^{o_2}$, so

$$\mu_{q_1 \otimes q_2, P_1 \otimes P_2} = \mu_{q_1, P_1} \otimes \mu_{q_2, P_2}. \quad (\text{PL})$$

Full revelation and no revelation. For $q \in \Delta(A)$, the no-information posterior law is δ_q . The full-revelation posterior law is

$$\chi_q := \sum_{a \in A} q(a) \delta_{e_a},$$

where e_a is the degenerate lottery on action a . No information leaves the observer's belief at the prior q ; full revelation identifies the realised action.

Axioms

The axioms are imposed directly on the family $\{\succeq_P\}_P$. They do not assume a posterior-law order, and they do not introduce a preference over pairs (q, P) . Cross-environment-looking statements are always written as comparisons between block-supported lotteries inside one fixed block environment.

The axioms treat traceable agency as a comparative property of an act lottery together with an observation procedure. They require stable rankings within each environment, coherent comparisons through labelled block environments, monotonicity when outcomes or act descriptions become less directly traceable, and disciplined behaviour under sequential observation. Disciplined behaviour under public randomisation is not assumed; it is derived in Lemma 6. Nor is any separability requirement across independent background components assumed: the form of separability that the representation argument needs is derived in Lemma 13.

(A1) Weak order and local non-triviality. For every finite channel $P : A \rightarrow \Delta(O)$, $\succeq_P$ is complete and transitive on $\Delta(A)$. The family is locally non-trivial: for every finite action set A with $|A| \geq 2$ and every full-support $q \in \Delta(A)$,

$$q^0 \succ_{\text{Id}_A \sqcup U_A} q^1,$$

where $\text{Id}_A : A \rightarrow \Delta(A)$ is full revelation and $U_A : A \rightarrow \Delta(\{*\})$ is the one-outcome uninformative channel.

This axiom gives each observation environment an ordinary weak order over act lotteries and fixes the orientation of the comparison. When the realised act is genuinely uncertain, observing it directly is strictly more traceable than observing nothing about it.

(A2) Continuity. For every pair of finite alphabets (A, O) , the set

$$\{(P, q, q') \in \Delta(O)^A \times \Delta(A) \times \Delta(A) : q \succeq_P q'\}$$

is closed. Thus within-environment rankings vary continuously with both the lotteries and the fixed channel.

Continuity says that traceability comparisons are stable under small changes in the procedure and the environment. Small modelling perturbations should not create abrupt reversals. The axiom is stated entirely in the primitives: it quantifies over channels and lotteries, not over derived Bayesian objects. Continuity of block comparisons in posterior-law convergence at a fixed full-support prior is not assumed; it is derived below (Lemma 7).

(A3) Block-comparison coherence. First, duplicating the same environment into two labelled blocks does not change the original ranking of lotteries: for every channel $P : A \rightarrow \Delta(O)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \iff q^0 \succeq_{P \sqcup P} (q')^1.$$

Second, block comparisons are invariant to adding or deleting irrelevant blocks. For every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff q_i^0 \succeq_{P_i \sqcup P_j} q_j^1.$$

Labelled blocks are a way to place procedures carried by different channels in a common comparison problem. The labels should not themselves create traceability, alter the ranking inside a duplicated environment, or let unused blocks affect the comparison between active blocks.

(A4) Outcome post-processing aversion. Let $P : A \rightarrow \Delta(O)$ and let $T : O \rightarrow \Delta(O')$ be stochastic. Then, for every $q \in \Delta(A)$,

$$q^0 \succeq_{P \sqcup PT} q^1.$$

Post-processing changes only the observed signal. Since a garbling can only discard or obscure information contained in the original outcome, the original procedure is weakly more traceable than its post-processed version.

(A5) Action-coarsening aversion. Let $S : A \rightarrow \Delta(A')$ be stochastic. For every $P : A \rightarrow \Delta(O)$, every $q \in \Delta(A)$, and every completion $\hat{P} : A' \rightarrow \Delta(O)$ of the Bayesian pushforward $S^q P$,

$$q^0 \succeq_{P \sqcup \hat{P}} (qS)^1.$$

When qS has full support, the completion is unique and we write it simply as S^qP ; otherwise, any use of S^qP in this axiom is understood for every completion. In particular, bijective relabelings are neutral by applying the axiom in both directions, using the corresponding completions on zero-probability rows if needed.

The act whose trace is evaluated may be reported through a possibly noisy description. The induced comparison is then between traceability of the original act and traceability of the described act generated from it. Such a description cannot display more observable trace than the act from which it is generated. Zero-probability rows of the described act are irrelevant to the comparison, no matter how they are completed.

(A6) Branchwise continuation monotonicity. Fix $q \in \Delta(A)$ and a first-stage channel $P_1 : A \rightarrow \Delta(O_1)$. Let $m(o) := m_{q,P_1}(o)$ and $r_o := r_o^{q,P_1}$ for all branches with $m(o) > 0$.

For each such branch, let $Q^o, R^o : A \rightarrow \Delta(O_2^o)$ be continuation channels. If

$$r_o^0 \succeq_{Q^o \sqcup R^o} r_o^1 \quad \text{for every } o \text{ with } m(o) > 0,$$

then

$$q^0 \succeq_{(P_1 \triangleright \{Q^o\}) \sqcup (P_1 \triangleright \{R^o\})} q^1.$$

If one branch comparison is strict and has positive probability, then the aggregate comparison is strict.

Sequential observation is evaluated branch by branch. After a first-stage signal, if one continuation leaves at least as much trace as another at every posterior that can be reached, then replacing the continuation branch by branch should not reduce overall traceability. If the improvement is strict on a reached branch, the aggregate comparison is strict.

No axiom is imposed on product environments. The product structure enters the argument only through Lemma 13 below, which shows that a statistically independent background component is behaviourally inert: it is derived from (A1), (A3)–(A6), because a product channel is a two-stage channel whose first stage reveals nothing about the foreground act.

Benchmark Verification

Before proving the representation theorem, we verify that the benchmark family of preferences generated by mutual information satisfies the axioms. This establishes consistency of the axiom system and gives the easy direction of the characterisation. The substantive content of the paper is the converse direction, proved in the sufficiency section.

Lemma 1 (Benchmark: mutual information satisfies the axioms). *For every finite channel $P : A \rightarrow \Delta(O)$, define*

$$q \succeq_P^I q' \iff I_{q,P}(A; O) \geq I_{q',P}(A; O).$$

Then the family $\{\succeq_P^I\}_P$ satisfies Axioms (A1)–(A6).

Proof. Write H_{Sh} for Shannon entropy on finite distributions, with $0 \log 0 := 0$, and abbreviate $I_{q,P}(A; O)$ to $I_{q,P}$ when the alphabets are clear. We repeatedly use the elementary identities

$$I_{q,P} = H_{\text{Sh}}(m_{q,P}) - \sum_{a \in A} q(a) H_{\text{Sh}}(P(\cdot | a)), \quad (\text{N1})$$

and

$$I_{q,P} = H_{\text{Sh}}(q) - \int_{\Delta(A)} H_{\text{Sh}}(r) d\mu_{q,P}(r). \quad (\text{N2})$$

Both formulae are read on the support of the relevant distributions; zero-probability rows and outcomes do not affect the value.

Axiom (A1). For each fixed P , the relation $\succeq_P^I$ is complete and transitive because it is represented by the real-valued function $q \mapsto I_{q,P}$. If $|A| \geq 2$ and q has full support, then full revelation gives

$$I_{q, \text{Id}_A} = H_{\text{Sh}}(q) > 0,$$

whereas the one-outcome uninformative channel U_A gives $I_{q, U_A} = 0$. Hence

$$q^0 \succ_{\text{Id}_A \sqcup U_A}^I q^1.$$

Axiom (A2). For fixed finite alphabets, (N1) shows that $(P, q) \mapsto I_{q,P}$ is continuous, since Shannon entropy is continuous on every finite simplex. Therefore the set

$$\{(P, q, q') : q \succeq_P^I q'\} = \{(P, q, q') : I_{q,P} \geq I_{q',P}\}$$

is closed. Posterior-law continuity of block comparisons is not an axiom; for the mutual-information family it also holds directly, by (N2) and continuity of H_{Sh} on the compact simplex $\Delta(A)$, consistently with its derivation from the axioms in Lemma 7 below.

Axiom (A3). For every channel P and lotteries $q, q' \in \Delta(A)$,

$$I_{q^0, P \sqcup P} = I_{q,P}, \quad I_{(q')^1, P \sqcup P} = I_{q',P},$$

because a block-supported lottery puts probability one on its own block and the block label is constant under that lottery. Hence

$$q \succeq_P^I q' \iff q^0 \succeq_{P \sqcup P}^I (q')^1.$$

The same observation proves invariance to adding or deleting irrelevant blocks. If q_i^i and q_j^j are supported on blocks i and j , their mutual-information values in $\bigsqcup_{k \in K} P_k$ are exactly

$$I_{q_i, P_i} \quad \text{and} \quad I_{q_j, P_j},$$

independently of all other blocks.

Axiom (A4). Let $T : O \rightarrow \Delta(O')$ be a stochastic outcome post-processing. Under PT , the variables satisfy the Markov chain

$$A \longrightarrow O \longrightarrow O'.$$

The data-processing inequality gives $I_{q,P}(A; O) \geq I_{q,PT}(A; O')$. Therefore

$$q^0 \succeq_{P \sqcup PT}^I q^1.$$

Axiom (A5). Let $S : A \rightarrow \Delta(A')$ be stochastic. Generate $A \sim q$, then A' from $S(\cdot | A)$, and O from $P(\cdot | A)$. Then A' is a garbling of A , and

$$A' \longleftarrow A \longrightarrow O$$

is a Markov fork. Data processing gives $I(A; O) \geq I(A'; O)$. The joint law of (A', O) is exactly the one induced by the Bayesian pushforward $S^q P$ on every row a' with $(qS)(a') > 0$. Values of a completion $\hat{P}$ on rows with $(qS)(a') = 0$ do not affect mutual information. Therefore

$$I_{q,P}(A; O) \geq I_{qS, \hat{P}}(A'; O)$$

for every completion $\hat{P}$ of $S^q P$, and so

$$q^0 \succeq_{P \sqcup \hat{P}}^I (qS)^1.$$

Axiom (A6). Fix a first-stage channel $P_1 : A \rightarrow \Delta(O_1)$. Let

$$m(o) = m_{q,P_1}(o), \quad r_o = r_o^{q,P_1}.$$

For a continuation profile $\mathcal{Q} = \{Q^o\}_o$, the chain rule for mutual information gives

$$I_{q,P_1 \triangleright \mathcal{Q}}(A; O_1, O_2) = I_{q,P_1}(A; O_1) + \sum_{o: m(o) > 0} m(o) I_{r_o, Q^o}(A; O_2^o). \quad (\text{N3})$$

Therefore, if

$$r_o^0 \succeq_{Q^o \sqcup R^o}^I r_o^1 \quad \text{for every positive-probability branch } o,$$

then $I_{r_o, Q^o} \geq I_{r_o, R^o}$ for every such o . Using (N3),

$$I_{q,P_1 \triangleright \{Q^o\}} \geq I_{q,P_1 \triangleright \{R^o\}},$$

so

$$q^0 \succeq_{(P_1 \triangleright \{Q^o\}) \sqcup (P_1 \triangleright \{R^o\})}^I q^1.$$

If one branch comparison is strict and that branch has positive probability, then the corresponding summand in (N3) is strictly larger, so the aggregate comparison is strict.

No product condition is imposed, so the verification ends here. For orientation, note that the mutual-information family satisfies the background-inertness property derived in Lemma 13 below, since for product priors and product channels

$$I_{q_1 \otimes q_2, P_1 \otimes P_2}(A_1, A_2; O_1, O_2) = I_{q_1, P_1}(A_1; O_1) + I_{q_2, P_2}(A_2; O_2), \quad (\text{N4})$$

so that the comparison of $P_1 \otimes R_2$ and $Q_1 \otimes R_2$ at $q_1 \otimes q_2$ reduces to the comparison of P_1 and Q_1 at q_1 , whatever the common background R_2 . Consistently with Lemma 13, this is a consequence of the chain rule (N3) and not an independent requirement. $\square$

Main Result

Theorem 2 (Mutual-information characterisation). *For a family $\{\succeq_P\}_P$ of weak orders over finite stochastic environments, the following are equivalent.*

- (i) *The family satisfies Axioms (A1)–(A6).*
- (ii) *For every finite channel $P : A \rightarrow \Delta(O)$ and every $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{q,P}(A; O) \geq I_{q',P}(A; O).$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$, $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff I_{q_i, P_i}(A_i; O_i) \geq I_{q_j, P_j}(A_j; O_j).$$

Proof. If (ii) holds, then $\{\succeq_P\}_P$ is the mutual-information family, so Lemma 1 gives (i). The sufficiency section proves (i) $\Rightarrow$ (ii), culminating in Lemma 22. For the moreover clause, apply (ii) to the fixed block environment $\bigsqcup_{k \in K} P_k$. A lottery supported on block i has constant block label, so

$$I_{q_i^i, \bigsqcup_{k \in K} P_k} = I_{q_i, P_i}(A_i; O_i),$$

and similarly for block j . $\square$

Proof of Sufficiency

Assume Axioms (A1)–(A6). We prove the implication (i) $\Rightarrow$ (ii) in Theorem 2. The proof is written first for full-support lotteries. Boundary cases are obtained by restricting to the support and then using continuity. The key point is that posterior-law sufficiency is not assumed; it is derived from block comparisons and outcome post-processing, and so is its continuity (Lemma 7).

Proof roadmap. Lemmas 3–4 derive posterior-law sufficiency from block comparisons and Blackwell equivalence. Lemma 6 derives public-coin independence from branchwise continuation monotonicity, before its first use. Lemma 7 derives continuity of block comparisons in posterior-law convergence from the primitive closedness of Axiom (A2). Lemmas 5–10 give fibrewise affine posterior-separable representations and basic invariances. Lemma 11 records undetectable-distinctions neutrality as a derived consequence of action coarsening, and Lemma 12 identifies comparisons at a boundary prior with comparisons on its support face. Lemma 13 then derives the only product-environment property the argument uses: an independent background component is behaviourally inert, because a product channel is a two-stage channel whose first stage reveals nothing about the foreground act. Lemma 14 chooses a coherent product gauge and proves quasi-additivity, allowing a possible bilinear interaction term. Lemma 16 moves cross-prior block comparisons into these unscaled values. Lemma 17 derives branch aggregation, and Lemma 18 derives the cocycle, the beta-form, and the normalised chain rule. Lemma 19 aligns the chain scales across support faces. Lemma 20 uses the chain rule and a two-grouping argument to eliminate the interaction term and prove the scale a_q is universal. Lemma 21 then proves entropy reduction, Faddeev recursion, and the mutual-information formula. Lemma 22 gives the fixed-environment representation.

Lemma 3 (Coherent block comparisons). *Under (A1) and (A3), pairwise comparisons between block-supported lotteries are independent of the block environment in which the two blocks are embedded. In particular, fix a finite action set A and a full-support $q \in \Delta(A)$. For channels P, Q, R on A , the comparisons*

$$q^0 \succeq_{P \sqcup Q} q^1$$

are complete and transitive as comparisons of channels at prior q . The same Axiom (A3) convention identifies the reversed block-supported comparison with its canonical two-block form:

$$q^1 \succeq_{P \sqcup Q} q^0 \iff q^0 \succeq_{Q \sqcup P} q^1.$$

Under (A2), these comparisons are also closed in the channels: if $P_n \rightarrow P$, $Q_n \rightarrow Q$, and

$$q^0 \succeq_{P_n \sqcup Q_n} q^1 \quad \text{for all } n,$$

then

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Proof. Completeness follows from Axiom (A1) applied to the two-block environment $P \sqcup Q$: either $q^0 \succeq_{P \sqcup Q} q^1$ or $q^1 \succeq_{P \sqcup Q} q^0$. In the second case, applying Axiom (A3) to the two-block environment with $P_0 = P$ and $P_1 = Q$, using the ordered pair $(i, j) = (1, 0)$, gives the canonical reverse comparison

$$q^1 \succeq_{P \sqcup Q} q^0 \iff q^0 \succeq_{Q \sqcup P} q^1.$$

For transitivity, suppose

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \text{and} \quad q^0 \succeq_{Q \sqcup R} q^1.$$

Form the three-block environment $P^0 \sqcup Q^1 \sqcup R^2$, and write q^P, q^Q, q^R for the lottery q supported on the P , Q , and R blocks. By the irrelevant-block clause of Axiom (A3) with block pair $(0, 1)$, the first comparison is equivalent to

$$q^P \succeq_{P^0 \sqcup Q^1 \sqcup R^2} q^Q,$$

and by the same clause with block pair $(1, 2)$, the second comparison is equivalent to

$$q^Q \succeq_{P^0 \sqcup Q^1 \sqcup R^2} q^R.$$

Transitivity of the weak order in this common environment gives

$$q^P \succeq_{P^0 \sqcup Q^1 \sqcup R^2} q^R.$$

Applying Axiom (A3) again, now deleting the irrelevant Q block, yields

$$q^0 \succeq_{P \sqcup R} q^1.$$

For closedness, the channel $P_n \sqcup Q_n$ converges to $P \sqcup Q$ in the finite-dimensional product simplex. Since the block-supported lotteries q^0, q^1 are fixed, Axiom (A2) gives

$$q^0 \succeq_{P \sqcup Q} q^1.$$

□

Lemma 4 (Prior-specific Blackwell equivalence). *Let $q \in \Delta(A)$ have full support. Let $P : A \rightarrow \Delta(O)$ and $Q : A \rightarrow \Delta(Y)$ be finite channels. If*

$$\mu_{q,P} = \mu_{q,Q},$$

then there are stochastic kernels $T : O \rightarrow \Delta(Y)$ and $T' : Y \rightarrow \Delta(O)$ such that

$$Q = PT, \quad P = QT'.$$

Proof. It is enough to construct T ; the construction of T' is symmetric. Let

$$m_P(o) := m_{q,P}(o), \quad m_Q(y) := m_{q,Q}(y).$$

If $m_P(o) = 0$, then $P(o | a) = 0$ for every $a \in A$, because q has full support. Such outcomes do not affect PT , so define the corresponding row of T arbitrarily. The same observation applies to null outcomes of Q .

Let $\mathcal{R}$ be the finite set of posterior vectors that occur with positive probability under either channel. For $r \in \mathcal{R}$, define

$$O_r := \{o \in O : m_P(o) > 0, r_o^{q,P} = r\}, \quad Y_r := \{y \in Y : m_Q(y) > 0, r_y^{q,Q} = r\}.$$

The sets $\{O_r\}_{r \in \mathcal{R}}$ partition the positive-probability outcomes of P , and the sets $\{Y_r\}_{r \in \mathcal{R}}$ partition the positive-probability outcomes of Q , by posterior vector. Equality of posterior laws implies that the total probability attached to posterior r is the same for the two channels:

$$\lambda_r := \sum_{o \in O_r} m_P(o) = \sum_{y \in Y_r} m_Q(y) > 0.$$

For $o \in O_r$ and $y \in Y_r$, set

$$T(y | o) := \frac{m_Q(y)}{\lambda_r},$$

and set $T(y | o) = 0$ when $o \in O_r$ and $y \notin Y_r$. This defines a stochastic row for every positive-probability outcome o , since the probabilities over Y_r sum to one. Rows corresponding to null outcomes of P were already chosen arbitrarily.

Fix $a \in A$ and $y \in Y_r$. Since all outcomes in O_r have posterior r , Bayes' rule gives

$$q(a)P(o | a) = m_P(o)r(a) \quad (o \in O_r).$$

Therefore

$$\sum_{o \in O_r} P(o | a) = \frac{\lambda_r r(a)}{q(a)}.$$

Using the definition of T ,

$$(PT)(y | a) = \sum_{o \in O_r} P(o | a) \frac{m_Q(y)}{\lambda_r} = \frac{m_Q(y)r(a)}{q(a)}.$$

But $y \in Y_r$ also has posterior r , so Bayes' rule for Q gives

$$Q(y | a) = \frac{m_Q(y)r(a)}{q(a)}.$$

Thus $(PT)(y | a) = Q(y | a)$ for every positive-probability y . If $m_Q(y) = 0$, then full support of q implies $Q(y | a) = 0$ for every a . No positive-probability outcome of P sends mass to such a y under the construction; rows of T indexed by null outcomes of P may be chosen arbitrarily, but those outcomes have $P(o | a) = 0$ for every a , so they contribute nothing to PT . Hence $(PT)(y | a) = 0 = Q(y | a)$ for every a , and therefore $Q = PT$. $\square$

Lemma 5 (Posterior-law sufficiency). *Fix a full-support $q \in \Delta(A)$. If $P : A \rightarrow \Delta(O)$ and $Q : A \rightarrow \Delta(Y)$ generate the same posterior law at q ,*

$$\mu_{q,P} = \mu_{q,Q},$$

then

$$q^0 \sim_{P \sqcup Q} q^1.$$

Consequently, by Lemma 3, at a fixed full-support prior the block comparison of channels depends only on the induced posterior law.

Proof. By Lemma 4, there are stochastic kernels T and T' such that $Q = PT$ and $P = QT'$. Axiom (A4) gives

$$q^0 \succeq_{P \sqcup PT} q^1, \quad q^0 \succeq_{Q \sqcup QT'} q^1.$$

Using $PT = Q$, the first comparison is

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Using $QT' = P$, the second comparison is

$$q^0 \succeq_{Q \sqcup P} q^1.$$

By the reverse-block equivalence in Lemma 3 (equivalently, Axiom (A3) applied to $P^0 \sqcup Q^1$ with ordered pair $(i, j) = (1, 0)$), this is equivalent to

$$q^1 \succeq_{P \sqcup Q} q^0.$$

Thus $q^0 \sim_{P \sqcup Q} q^1$. $\square$

Lemma 6 (Derived public-coin independence). *Assume Axioms (A1), (A3), (A4), and (A6). Fix a finite action set A , a full-support $q \in \Delta(A)$, and channels P, Q, R on A . For every $\lambda \in (0, 1)$,*

$$q^0 \succeq_{P \sqcup Q} q^1 \iff q^0 \succeq_{(\lambda P \oplus (1-\lambda)R) \sqcup (\lambda Q \oplus (1-\lambda)R)} q^1.$$

Proof. Throughout, for channels V, W on A write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$, and $V \succ_q W$ for the corresponding strict comparison. By Lemma 3 the comparison $\succeq_q$ is complete and transitive, and by Lemma 5 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$. In particular, channels with equal posterior laws at q may be substituted on either side of any comparison; we refer to this as *substitution*. Substitution also preserves strict comparisons: if $\mu_{q,V} = \mu_{q,V'}$ and $\mu_{q,W} = \mu_{q,W'}$, then $V \sim_q V'$ and $W \sim_q W'$ by Lemma 5, and transitivity of the complete relation $\succeq_q$ gives $V \succ_q W$ if and only if $V' \succ_q W'$.

Step 1: the public coin is an uninformative first stage. Define the constant channel $C : A \rightarrow \Delta(\{0, 1\})$ by

$$C(0 \mid a) := \lambda, \quad C(1 \mid a) := 1 - \lambda \quad (a \in A).$$

Since $\lambda \in (0, 1)$, both branches have positive mass: $m_{q,C}(0) = \lambda$ and $m_{q,C}(1) = 1 - \lambda$. Because the rows of C are constant, Bayes' rule gives, for each $o \in \{0, 1\}$,

$$r_o^{q,C}(a) = \frac{q(a)C(o \mid a)}{m_{q,C}(o)} = q(a),$$

so both branch posteriors equal the prior: $r_0 = r_1 = q$.

Step 2: padding the continuations to a common alphabet. Axiom (A6) requires the two continuation channels on a given branch to share an outcome alphabet, whereas P and Q may not. Let $O_{PQ} := O_P \sqcup O_Q$ and define $\tilde{P}, \tilde{Q} : A \rightarrow \Delta(O_{PQ})$ by

$$\tilde{P}(z \mid a) := \begin{cases} P(z \mid a) & z \in O_P, \\ 0 & z \in O_Q, \end{cases} \quad \tilde{Q}(z \mid a) := \begin{cases} 0 & z \in O_P, \\ Q(z \mid a) & z \in O_Q. \end{cases}$$

Padding adds only zero-probability outcomes, so the positive-mass outcomes and their posteriors are unchanged: $\mu_{q,\tilde{P}} = \mu_{q,P}$ and $\mu_{q,\tilde{Q}} = \mu_{q,Q}$. By substitution,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff \tilde{P} \succeq_q \tilde{Q},$$

and the same equivalence holds for the strict comparisons.

Step 3: the compounds implement the public-coin mixtures. Consider the two-stage channels with first stage C , continuation pair $\tilde{P}$ versus $\tilde{Q}$ on branch 0, and the common continuation R on branch 1:

$$C \triangleright \{\tilde{P}, R\} \quad \text{and} \quad C \triangleright \{\tilde{Q}, R\},$$

where the profile notation lists the branch-0 and branch-1 continuations in order. By the definition of two-stage composition,

$$(C \triangleright \{\tilde{P}, R\})(0, z \mid a) = \lambda \tilde{P}(z \mid a), \quad (C \triangleright \{\tilde{P}, R\})(1, z \mid a) = (1 - \lambda) R(z \mid a).$$

An outcome $(0, z)$ has marginal mass $\lambda m_{q,\tilde{P}}(z)$, and when this is positive the constant factor λ cancels in Bayes' rule, so its posterior is $r_z^{q,\tilde{P}}$; symmetrically for outcomes $(1, z)$. Hence

$$\mu_{q,C \triangleright \{\tilde{P}, R\}} = \lambda \mu_{q,\tilde{P}} + (1 - \lambda) \mu_{q,R} = \lambda \mu_{q,P} + (1 - \lambda) \mu_{q,R} = \mu_{q,\lambda P \oplus (1-\lambda)R},$$

using the public-coin mixing identity, and likewise $\mu_{q,C \triangleright \{\tilde{Q}, R\}} = \mu_{q,\lambda Q \oplus (1-\lambda)R}$. By substitution, the right-hand comparison of the lemma is therefore equivalent to

$$C \triangleright \{\tilde{P}, R\} \succeq_q C \triangleright \{\tilde{Q}, R\}. \quad (*)$$

Step 4: forward implication. Assume $q^0 \succeq_{P \sqcup Q} q^1$. We verify the branch hypotheses of Axiom (A6) for the first stage C and the continuation profiles $\{\tilde{P}, R\}$ versus $\{\tilde{Q}, R\}$. Both branches have positive mass and both branch posteriors equal q (Step 1). On branch 0 the hypothesis is $q^0 \succeq_{\tilde{P} \sqcup \tilde{Q}} q^1$, which holds by Step 2. On branch 1 the two profiles share the continuation R , and the hypothesis $q^0 \succeq_{R \sqcup R} q^1$ holds because Axiom (A1) gives $q \succeq_R q$ and the duplication clause of Axiom (A3) converts it into the two-block form. Axiom (A6) then yields (*), hence the mixture comparison by Step 3.

Step 5: converse, via the strictness clause. Assume the mixture comparison holds, so that (*) holds by Step 3, and suppose towards a contradiction that $q^0 \succeq_{P \sqcup Q} q^1$ fails. By completeness and the reverse-block equivalence of Lemma 3, $Q \succ_q P$, and Step 2 gives $\tilde{Q} \succ_q \tilde{P}$. Now apply Axiom (A6) with the roles of the two profiles exchanged: the branch-0 comparison $\tilde{Q} \succ_q \tilde{P}$ is strict on a branch of mass $\lambda > 0$, and the branch-1 comparison $R \succeq_q R$ holds as in Step 4. The strictness clause of Axiom (A6) yields

$$C \triangleright \{\tilde{Q}, R\} \succ_q C \triangleright \{\tilde{P}, R\},$$

which contradicts (*). Hence $q^0 \succeq_{P \sqcup Q} q^1$. $\square$

The lemma shows that public-coin independence is not an independent behavioural commitment: a public coin is exactly a first-stage observation that reveals nothing about the act, so invariance under common public mixtures is branchwise continuation monotonicity applied through trivial branch posteriors. The statement is given at full-support priors, which is the only case used below (Lemma 8).

Lemma 7 (Posterior-law continuity). *Fix a full-support $q \in \Delta(A)$. Let P_n, Q_n, P, Q be finite channels on A with*

$$\mu_{q, P_n} \Rightarrow \mu_{q, P}, \quad \mu_{q, Q_n} \Rightarrow \mu_{q, Q}.$$

If

$$q^0 \succeq_{P_n \sqcup Q_n} q^1 \quad \text{for all } n,$$

then

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Proof. Throughout the proof, for channels V, W on A write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$. By Lemma 3 this comparison is complete and transitive, and by Lemma 5 it depends only on the pair of posterior laws $(\mu_{q, V}, \mu_{q, W})$; in particular, channels with equal posterior laws at q may be substituted on either side of any comparison. We refer to this as *substitution*.

For a finite-support Bayes-plausible law

$$\mu = \sum_{k \in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

with atoms listed so that $\lambda_k > 0$, define the canonical implementing channel $C_\mu : A \rightarrow \Delta(K)$ by

$$C_\mu(k \mid a) := \frac{\lambda_k r_k(a)}{q(a)}.$$

Full support of q and $\sum_k \lambda_k r_k = q$ make C_μ a channel with $\mu_{q, C_\mu} = \mu$.

Step 1: merging posteriors moves weakly down. Let $\mu = \sum_{k \in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q$ and let $\{K_1, \dots, K_m\}$ partition K into nonempty groups. The merged law is

$$\text{mrg}(\mu) := \sum_{j=1}^m w_j \delta_{s_j}, \quad w_j := \sum_{k \in K_j} \lambda_k, \quad s_j := \frac{1}{w_j} \sum_{k \in K_j} \lambda_k r_k.$$

Barycentres are preserved, so $\text{mrg}(\mu) \in \mathcal{M}_q$. Let $T : K \rightarrow \Delta(\{1, \dots, m\})$ be the deterministic kernel with $T(j \mid k) = 1$ if and only if $k \in K_j$. Then

$$(C_\mu T)(j \mid a) = \sum_{k \in K_j} \frac{\lambda_k r_k(a)}{q(a)} = \frac{w_j s_j(a)}{q(a)} = C_{\text{mrg}(\mu)}(j \mid a),$$

so Axiom (A4) gives

$$C_\mu \succeq_q C_{\text{mrg}(\mu)}.$$

Conversely, say that $\mu' \in \mathcal{M}_q$ is a *spread* of μ if μ' is obtained by replacing each atom (λ_k, r_k) of μ by finitely many atoms $(\lambda_k w_{kj}, v_{kj})_j$ with $w_{kj} \geq 0$, $\sum_j w_{kj} = 1$, and $\sum_j w_{kj} v_{kj} = r_k$. Then $\mu = \text{mrg}(\mu')$ for the grouping by k , so

$$C_{\mu'} \succeq_q C_\mu.$$

Atoms with $w_{kj} = 0$ are deleted from the list; and if the same posterior appears in several list entries, any two listings of the same measure induce the same posterior law at q , so the corresponding canonical channels are interchangeable by substitution. Merging outcomes can only obscure which act was realised; splitting an atom into a barycentric family is the reverse operation.

Step 2: bounded supports. We claim the lemma holds under the additional hypothesis that all laws μ_{q, P_n} and μ_{q, Q_n} have support of size at most K , for some fixed K . Write

$$\mu_{q, P_n} = \sum_{k=1}^K \lambda_{n,k} \delta_{r_{n,k}},$$

padding with zero-mass atoms ($\lambda_{n,k} = 0$, $r_{n,k} := q$) so that exactly K terms appear, and define the K -outcome channel $C_n : A \rightarrow \Delta(\{1, \dots, K\})$ by

$$C_n(k \mid a) := \frac{\lambda_{n,k} r_{n,k}(a)}{q(a)}.$$

Each row sums to one because $\sum_k \lambda_{n,k} r_{n,k} = q$, zero-mass columns contribute nothing, and $\mu_{q, C_n} = \mu_{q, P_n}$. Define D_n from μ_{q, Q_n} in the same way. By substitution,

$$C_n \succeq_q D_n \quad \text{for all } n.$$

The parameter sequences of C_n and D_n jointly live in the compact metric space $([0, 1] \times \Delta(A))^{2K}$; pass to a single subsequence along which $\lambda_{n,k} \rightarrow \lambda_k^*$ and $r_{n,k} \rightarrow r_k^*$ for every k , simultaneously for both channels. Then $C_n \rightarrow C^*$ entrywise, where $C^*(k \mid a) = \lambda_k^* r_k^*(a)/q(a)$; each row of C^* sums to one as an entrywise limit of rows summing to one, so C^* is a channel. For every continuous f on $\Delta(A)$,

$$\int f d\mu_{q, P_n} = \sum_{k=1}^K \lambda_{n,k} f(r_{n,k}) \longrightarrow \sum_{k=1}^K \lambda_k^* f(r_k^*),$$

so by uniqueness of weak limits

$$\mu_{q, P} = \sum_{k=1}^K \lambda_k^* \delta_{r_k^*} = \mu_{q, C^*},$$

where the second equality holds because outcome k of C^* has probability λ_k^* and, when $\lambda_k^* > 0$, posterior r_k^* . Similarly $D_n \rightarrow D^*$ entrywise with $\mu_{q, D^*} = \mu_{q, Q}$. The closedness clause of Lemma 3 applied along the subsequence gives

$$C^* \succeq_q D^*,$$

and substitution returns $q^0 \succeq_{P \sqcup Q} q^1$. This proves the bounded-support case; note that only the primitive closedness of Axiom (A2) has been used, through Lemma 3.

Step 3: the general case, by an ε -grid sandwich. Write $\mu_n := \mu_{q, P_n}$, $\nu_n := \mu_{q, Q_n}$, $\mu := \mu_{q, P}$, $\nu := \mu_{q, Q}$. The only remaining gap is that the supports of μ_n, ν_n need not be bounded. Fix $\varepsilon > 0$. Two finite partitions of $\Delta(A)$ are used, one for each surrogate; they play independent roles and need not be related. For the spread, fix a finite triangulation $\mathcal{T}$ of $\Delta(A)$ into simplices of diameter at most ε , together with a measurable partition $\{\tilde{S} : S \in \mathcal{T}\}$ of $\Delta(A)$ with $\tilde{S} \subseteq S$, assigning each shared face to exactly one incident cell. For the merge, fix a finite Borel partition $\{E_1, \dots, E_M\}$ of $\Delta(A)$ into sets of diameter at most ε whose relative boundaries carry no ν -mass: cover $\Delta(A)$ by finitely many open balls of radius at most $\varepsilon/2$, choosing the radii so that no atom of ν lies on a bounding sphere (ν has finitely many atoms, so all but finitely many radii work), and disjointify.

For finite-support $\rho \in \mathcal{M}_q$ define two bounded-support surrogates. Both remain in $\mathcal{M}_q$: the spread operation replaces each posterior by a barycentric decomposition of itself, while the merge operation replaces a group of posteriors by their conditional mean, weighted by the total mass of the group. Cells with zero mass are ignored (or, equivalently, padded by an arbitrary posterior with zero weight when a fixed outcome alphabet is convenient). The *grid spread* $\text{sp}_\varepsilon(\rho)$ replaces each atom (λ, β) with $\beta \in \tilde{S}$ by the atoms $(\lambda w_v(\beta), v)_v$, where v ranges over the vertices of S and $(w_v(\beta))_v$ are the barycentric coordinates of β in S . It is a spread of ρ in the sense of Step 1, is supported on the fixed vertex set of $\mathcal{T}$, and satisfies $C_{\text{sp}_\varepsilon(\rho)} \succeq_q C_\rho$. The *cell merge* $\text{mg}_\varepsilon(\rho)$ merges the atoms of ρ within each class E_j of the Borel partition; it has at most one atom per class and satisfies $C_\rho \succeq_q C_{\text{mg}_\varepsilon(\rho)}$ by Step 1, since Step 1 allows any grouping of atoms.

By substitution and transitivity (Lemma 3), the hypothesis $P_n \succeq_q Q_n$ gives the chain

$$C_{\text{sp}_\varepsilon(\mu_n)} \succeq_q C_{\mu_n} \succeq_q C_{\nu_n} \succeq_q C_{\text{mg}_\varepsilon(\nu_n)}.$$

Both surrogate sequences converge weakly on fixed alphabets. For the spread side, let f be continuous on $\Delta(A)$. The triangulation defines a continuous piecewise-affine interpolant $I_{\mathcal{T}}f$ by

$$I_{\mathcal{T}}f(r) := \sum_{v \in \text{vert}(S)} w_v(r) f(v) \quad \text{for } r \in S.$$

This is well-defined: on a shared face the barycentric coordinates computed in the two incident simplices put zero weight on vertices outside the face and coincide with the intrinsic barycentric coordinates on that face. Hence $I_{\mathcal{T}}f$ is continuous on all of $\Delta(A)$. Moreover, for every finite-support ρ ,

$$\int f d\text{sp}_\varepsilon(\rho) = \int I_{\mathcal{T}}f d\rho,$$

because each posterior is replaced by the barycentric lottery over the vertices of its simplex. Thus $\mu_n \Rightarrow \mu$ gives $\text{sp}_\varepsilon(\mu_n) \Rightarrow \text{sp}_\varepsilon(\mu)$. For the merge side, each class E_j has relative boundary contained in the chosen bounding spheres, which carry no ν -mass, so E_j is a ν -continuity set; therefore $\nu_n(E_j) \rightarrow \nu(E_j)$ and, since the truncated map $r \mapsto r \mathbf{1}_{E_j}(r)$ is bounded and its discontinuity set lies in the relative boundary of E_j , which is ν -null, $\int_{E_j} r d\nu_n \rightarrow \int_{E_j} r d\nu$ for every class. Classes with $\nu(E_j) > 0$ thus have converging merged masses and merged locations. If $\nu(E_j) = 0$, then the merged mass tends to zero, so the chosen location in that class is immaterial for weak convergence; padding such a class by any fixed posterior only changes a zero-weight atom. Hence $\text{mg}_\varepsilon(\nu_n) \Rightarrow \text{mg}_\varepsilon(\nu)$.

The laws $\text{sp}_\varepsilon(\mu_n)$ and $\text{mg}_\varepsilon(\nu_n)$ have supports of size at most $N(\varepsilon)$, the maximum of the number of vertices of $\mathcal{T}$ and the number M of classes of the Borel partition. Step 2 applied to the chain's endpoints therefore yields

$$C_{\text{sp}_\varepsilon(\mu)} \succeq_q C_{\text{mg}_\varepsilon(\nu)}.$$

Finally, send $\varepsilon \rightarrow 0$ along a sequence $\varepsilon_m \downarrow 0$ with triangulations and Borel partitions chosen as above. The law $\text{sp}_{\varepsilon_m}(\mu)$ has at most $k_P |A|$ atoms, where k_P is the number of atoms of μ : each atom splits among the $|A|$ vertices of one cell. Each new atom lies within ε_m of the atom it came from and group masses are preserved, so $\text{sp}_{\varepsilon_m}(\mu) \Rightarrow \mu$. The law $\text{mg}_{\varepsilon_m}(\nu)$ has at most k_Q atoms, where k_Q is the number of atoms of ν : only classes meeting the support of ν carry mass. Once ε_m is smaller than the minimal distance between distinct atoms of ν , each class contains at most one atom of ν and $\text{mg}_{\varepsilon_m}(\nu) = \nu$ exactly. The comparisons $C_{\text{sp}_{\varepsilon_m}(\mu)} \succeq_q C_{\text{mg}_{\varepsilon_m}(\nu)}$ hold for every m , the supports are uniformly bounded, and the laws converge weakly to μ and ν ; one further application of Step 2 gives $C_\mu \succeq_q C_\nu$. Substitution returns

$$q^0 \succeq_{P \sqcup Q} q^1. \quad \square$$

The lemma shows that continuity of block comparisons in posterior-law convergence is not an independent behavioural commitment: it is the primitive closedness of Axiom (A2) seen through the quotient that Axioms (A3) and (A4) impose, with the convex order supplying compactness where outcome alphabets grow without bound. Only weak preference is closed in the limit; no strictness claim is made or used.

Lemma 8 (Posterior-separable representation). *Fix a full-support $q \in \Delta(A)$. There is a continuous affine functional*

$$F_q : \mathcal{M}_q \rightarrow \mathbb{R}$$

such that, for any two channels P, Q on A ,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

Moreover, for some continuous $\phi_q : \Delta(A) \rightarrow \mathbb{R}$,

$$F_q(\mu) = \int_{\Delta(A)} \phi_q(r) d\mu(r),$$

and therefore

$$F_q(\mu_{q,P}) = \sum_{o: m_{q,P}(o) > 0} m_{q,P}(o) \phi_q(r_o^{q,P}).$$

The affine functional is unique up to positive affine transformations. The integrand ϕ_q is not pointwise unique; adding an affine function of r leaves its integral on the fibre $\mathcal{M}_q$ changed only by a constant.

Proof. First note that every finite-support Bayes-plausible law is implemented by a finite channel. If

$$\mu = \sum_{k=1}^n \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

define

$$P(k | a) = \frac{\lambda_k r_k(a)}{q(a)}.$$

Since q has full support and $\sum_k \lambda_k r_k = q$, this is a channel. Its outcome probability at k is λ_k , and the posterior after k is r_k ; hence $\mu_{q,P} = \mu$.

Define a comparison on $\mathcal{M}_q$ as follows: for $\mu, \nu \in \mathcal{M}_q$, choose any channels P, Q implementing them and compare the two block-supported copies of q in $P \sqcup Q$. This comparison is independent of the chosen implementations. Indeed, if P, P' both implement μ and Q, Q' both implement ν , then Lemma 5 gives

$$q^0 \sim_{P \sqcup P'} q^1, \quad q^0 \sim_{Q \sqcup Q'} q^1.$$

Placing the relevant blocks in common block environments and using Lemma 3, transitivity gives

$$q^0 \succeq_{P \sqcup Q} q^1 \iff q^0 \succeq_{P' \sqcup Q'} q^1.$$

Thus the quotient comparison on $\mathcal{M}_q$ is well defined; denote it by $\succeq_q$. This construction is only at the fixed full-support prior q , and every law in $\mathcal{M}_q$ is finite-support and, by the preceding construction, implemented by a finite channel. Lemma 3 also gives completeness and transitivity of this quotient comparison.

The set $\mathcal{M}_q$ is a mixture space under ordinary convex combination. It is convex because barycentres are preserved by convex combination. The quotient relation $\succeq_q$ is a weak order by Lemma 3. Public-coin mixtures of implementing channels correspond exactly to convex mixtures of posterior laws:

$$\mu_{q, \lambda P \oplus (1-\lambda)R} = \lambda \mu_{q,P} + (1-\lambda) \mu_{q,R}.$$

Therefore the derived public-coin independence property (Lemma 6) gives mixture independence: for every $\mu, \nu, \rho \in \mathcal{M}_q$ and $\lambda \in (0, 1)$,

$$\mu \succeq_q \nu \iff \lambda \mu + (1-\lambda) \rho \succeq_q \lambda \nu + (1-\lambda) \rho.$$

The relative weak topology on $\mathcal{M}_q$ is inherited from the weak topology on probability measures over the compact metric simplex $\Delta(A)$, and convex-mixture paths are continuous in this topology. Lemma 7 gives relatively closed upper and lower contour sets. Indeed, if $\mu_n \Rightarrow \mu$ with $\mu_n, \mu, z \in \mathcal{M}_q$ and $\mu_n \succeq_q z$, choose finite implementing channels for these laws and apply Lemma 7 (with the constant sequence implementing z) to get $\mu \succeq_q z$; the argument for $\{\mu : z \succeq_q \mu\}$ is the same. Therefore, for fixed $\mu, \nu, \rho \in \mathcal{M}_q$, the sets

$$\{\lambda \in [0, 1] : \lambda \mu + (1-\lambda) \nu \succeq_q \rho\} \quad \text{and} \quad \{\lambda \in [0, 1] : \rho \succeq_q \lambda \mu + (1-\lambda) \nu\}$$

are closed.

By the Herstein–Milnor mixture-space theorem [27], there is a mixture-preserving representation $F_q : \mathcal{M}_q \rightarrow \mathbb{R}$:

$$\mu \succeq_q \nu \iff F_q(\mu) \geq F_q(\nu),$$

and

$$F_q(\lambda \mu + (1-\lambda) \nu) = \lambda F_q(\mu) + (1-\lambda) F_q(\nu).$$

If the quotient order is nonconstant, this representative is unique up to positive affine transformations. If the quotient order is constant, take $F_q \equiv 0$.

Moreover, F_q is continuous in the relative weak topology. For any $z \in \mathcal{M}_q$, the sets

$$\{\mu : F_q(\mu) \geq F_q(z)\} = \{\mu : \mu \succeq_q z\} \quad \text{and} \quad \{\mu : F_q(\mu) \leq F_q(z)\} = \{\mu : z \succeq_q \mu\}$$

are relatively closed by the preceding Lemma 7 argument. Since F_q is affine, its range on the convex set $\mathcal{M}_q$ is an interval. Hence every upper and lower level set of F_q is relatively closed: levels outside the range are trivial, and levels inside the range equal $F_q(z)$ for some $z \in \mathcal{M}_q$. Thus F_q is both upper and lower semicontinuous, and therefore continuous. For any channels P, Q on A ,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

It remains to write the affine functional in posterior-separable form, without leaving the finite-support domain. Let $X := \Delta(A)$. Choose

$$0 < \varepsilon < \min_{a \in A} q(a).$$

For each $r \in X$, define

$$s_r := \frac{q - \varepsilon r}{1 - \varepsilon} \in \Delta(A)$$

and

$$\eta_r := \varepsilon \delta_r + (1 - \varepsilon) \sum_{a \in A} s_r(a) \delta_{e_a}.$$

Then $\eta_r \in \mathcal{M}_q$, because its barycentre is

$$\varepsilon r + (1 - \varepsilon) s_r = q.$$

Choose constants c_a such that

$$\sum_{a \in A} q(a) c_a = F_q(\chi_q),$$

for instance $c_a = F_q(\chi_q)$ for every a . Define

$$\phi_q(r) := \frac{F_q(\eta_r) - (1 - \varepsilon) \sum_{a \in A} s_r(a) c_a}{\varepsilon}.$$

The map $r \mapsto \eta_r$ is weakly continuous and F_q is continuous, so ϕ_q is continuous.

Now take any finite-support law

$$\mu = \sum_{k=1}^n \lambda_k \delta_{r_k} \in \mathcal{M}_q.$$

Since $\sum_k \lambda_k r_k = q$, we have

$$\sum_{k=1}^n \lambda_k s_{r_k} = q.$$

Therefore

$$\sum_{k=1}^n \lambda_k \eta_{r_k} = \varepsilon \mu + (1 - \varepsilon) \chi_q.$$

By affinity of F_q ,

$$\sum_{k=1}^n \lambda_k F_q(\eta_{r_k}) = F_q(\varepsilon \mu + (1 - \varepsilon) \chi_q) = \varepsilon F_q(\mu) + (1 - \varepsilon) F_q(\chi_q).$$

Using $\sum_k \lambda_k s_{r_k} = q$ and $\sum_a q(a) c_a = F_q(\chi_q)$ gives

$$\sum_{k=1}^n \lambda_k \phi_q(r_k) = F_q(\mu).$$

This is exactly

$$F_q(\mu) = \int_X \phi_q(r) d\mu(r)$$

for every finite-support $\mu \in \mathcal{M}_q$. Applying this to $\mu_{q,P}$ gives the stated finite-sum formula. The affine representative F_q is unique up to positive affine transformations by Herstein–Milnor. The integrand is not pointwise unique on a fixed barycentre fibre: if ℓ is affine in r , then $\int \ell(r) d\mu(r) = \ell(q)$ for every $\mu \in \mathcal{M}_q$. $\square$

Lemma 9 (Refinement monotonicity, convexity, and normalisation). *Fix a full-support $q \in \Delta(A)$ and let F_q, ϕ_q be as in Lemma 8. If a finite law $\mu \in \mathcal{M}_q$ is obtained from another finite law $\nu \in \mathcal{M}_q$ by replacing each posterior of ν by a finite Bayes-plausible continuation experiment, then*

$$F_q(\mu) \geq F_q(\nu).$$

Consequently, ϕ_q may be chosen convex. After replacing ϕ_q by an observationally equivalent representative and subtracting a constant from F_q , one may impose

$$\phi_q(q) = F_q(\delta_q) = 0, \quad \phi_q(r) \geq 0 \quad \text{for every } r \in \Delta(A),$$

and therefore

$$F_q(\mu) \geq 0 \quad \text{for every } \mu \in \mathcal{M}_q.$$

Proof. Write

$$\nu = \sum_{i=1}^n m_i \delta_{r_i}, \quad \mu = \sum_{i=1}^n m_i \nu_i, \quad \nu_i \in \mathcal{M}_{r_i}.$$

Choose a channel $P : A \rightarrow \Delta(\{1, \dots, n\})$ implementing ν at prior q . For each i , choose a finite continuation channel $Q^i : A \rightarrow \Delta(Z_i)$ implementing ν_i at prior r_i ; on actions outside the support of r_i , define the rows arbitrarily. Those rows are never reached under prior r_i and do not affect ν_i ; moreover, if $r_i(a) = 0$, then $q(a)P(i | a) = m_i r_i(a) = 0$, so they do not affect the compound posterior law at q . The compound channel

$$C := P \triangleright \{Q^i\}_{i=1}^n$$

implements μ at prior q . Let $T : \bigsqcup_{i=1}^n (\{i\} \times Z_i) \rightarrow \Delta(\{1, \dots, n\})$ be the kernel $T(i | (i, z)) = 1$; then $CT = P$. Axiom (A4) applied to C and T gives

$$q^0 \succeq_{C \sqcup P} q^1.$$

By Lemma 8,

$$F_q(\mu) \geq F_q(\nu).$$

To prove convexity, take $r, s_1, \dots, s_k \in \Delta(A)$ with

$$r = \sum_{j=1}^k \lambda_j s_j, \quad \lambda_j \geq 0, \quad \sum_j \lambda_j = 1.$$

Choose $0 < \varepsilon < \min_a q(a)$ and set

$$t := \frac{q - \varepsilon r}{1 - \varepsilon}.$$

This belongs to $\Delta(A)$: for each a , $q(a) - \varepsilon r(a) \geq q(a) - \varepsilon > 0$, and the coordinates sum to one. The law

$$\varepsilon \sum_{j=1}^k \lambda_j \delta_{s_j} + (1 - \varepsilon) \delta_t$$

is a refinement of

$$\varepsilon \delta_r + (1 - \varepsilon) \delta_t,$$

and both belong to $\mathcal{M}_q$. Refinement monotonicity and the integral representation give

$$\varepsilon \sum_{j=1}^k \lambda_j \phi_q(s_j) + (1 - \varepsilon) \phi_q(t) \geq \varepsilon \phi_q(r) + (1 - \varepsilon) \phi_q(t).$$

Cancelling the common term and dividing by ε yields

$$\phi_q(r) \leq \sum_{j=1}^k \lambda_j \phi_q(s_j),$$

so ϕ_q is convex.

Since q is in the relative interior of the finite-dimensional simplex, the supporting-hyperplane theorem for finite convex functions gives a relative subgradient $g \in \mathbb{R}^A$ of the continuous convex function ϕ_q at q . Define

$$\tilde{\phi}_q(r) := \phi_q(r) - g \cdot (r - q).$$

For every $\mu \in \mathcal{M}_q$,

$$\int g \cdot (r - q) d\mu(r) = g \cdot (b(\mu) - q) = 0,$$

so $\tilde{\phi}_q$ represents the same affine functional on $\mathcal{M}_q$. The subgradient inequality gives

$$\tilde{\phi}_q(r) \geq \tilde{\phi}_q(q) \quad \text{for all } r \in \Delta(A).$$

Finally replace

$$F_q(\mu) \quad \text{by} \quad F_q(\mu) - F_q(\delta_q)$$

and replace $\tilde{\phi}_q$ by

$$\tilde{\phi}_q - \tilde{\phi}_q(q).$$

Before this final constant shift, the representation with $\tilde{\phi}_q$ gives

$$F_q(\delta_q) = \int \tilde{\phi}_q(r) d\delta_q(r) = \tilde{\phi}_q(q).$$

Thus the same constant is subtracted from both sides of the integral representation, and subtracting a constant from F_q preserves all ordinal comparisons on $\mathcal{M}_q$. The final normalisation gives

$$\phi_q(q) = F_q(\delta_q) = 0,$$

and makes ϕ_q non-negative on the whole simplex. Hence

$$F_q(\mu) = \int \phi_q(r) d\mu(r) \geq 0$$

for every $\mu \in \mathcal{M}_q$. □

Lemma 10 (Cross-fibre ordinal equivalence). *Action coarsening implies ordinal cross-fibre equivalence for bijections. In particular, all no-information experiments have the same normalised value zero, and for every bijection $\pi : A \rightarrow A'$ and full-support $q \in \Delta(A)$, there exists $c_{\pi,q} > 0$ such that*

$$F_{q\pi}(\mu\pi) = c_{\pi,q} F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$. The exact equality $c_{\pi,q} = 1$ for every q will be derived as Corollary 15 after product normalisation is established.

Proof. Ordinal equivalence. Let $\pi : A \rightarrow A'$ be a bijection. View π as the deterministic kernel $S(a' | a) = \mathbf{1}_{a'=\pi(a)}$. For any channel $P : A \rightarrow \Delta(O)$ and full-support $q \in \Delta(A)$, the Bayesian pushforward satisfies $S^q P = P\pi^{-1}$ (explicitly: $(S^q P)(o | a') = P(o | \pi^{-1}(a'))$). Axiom (A5) gives

$$q^0 \succeq_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

For the reverse inequality, apply Axiom (A5) to the bijection $\pi^{-1} : A' \rightarrow A$, the prior $q\pi \in \Delta(A')$, and the channel $P\pi^{-1} : A' \rightarrow \Delta(O)$. Then $(\pi^{-1})^{q\pi}(P\pi^{-1}) = P$ (the inverse bijection undoes the relabeling), and $(q\pi)\pi^{-1} = q$. Hence

$$(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1.$$

By the second part of Axiom (A3), block comparisons depend only on the pair of channels involved, not on numeric labels. The comparison $(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1$ asks whether $(q\pi, P\pi^{-1})$ is at least as good as (q, P) . Swapping the block labels (so that P becomes block 0 and $P\pi^{-1}$ becomes block 1) gives the equivalent statement $q^0 \preceq_{P \sqcup P\pi^{-1}} (q\pi)^1$. Combining with the first inequality gives

$$q^0 \sim_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

If P implements $\mu \in \mathcal{M}_q$, then $P\pi^{-1}$ implements $\mu\pi \in \mathcal{M}_{q\pi}$: for every positive-probability outcome o ,

$$m_{q\pi, P\pi^{-1}}(o) = m_{q, P}(o), \quad r_o^{q\pi, P\pi^{-1}} = r_o^{q, P} \pi.$$

Now take arbitrary $\mu, \nu \in \mathcal{M}_q$, and choose channels P, Q implementing them at prior q . Let E be the four-block environment with blocks $P, P\pi^{-1}, Q$, and $Q\pi^{-1}$. Write x, x', y, y' for the block-supported lotteries $q, q\pi, q$, and $q\pi$ on these four blocks, respectively. By Lemma 3, the two two-block indifferences give $x \sim_E x'$ and $y \sim_E y'$. Hence, by transitivity of the weak order on E ,

$$x \succeq_E y \iff x' \succeq_E y'.$$

Applying Lemma 3 again to delete irrelevant blocks gives

$$q^0 \succeq_{P \sqcup Q} q^1 \iff (q\pi)^0 \succeq_{P\pi^{-1} \sqcup Q\pi^{-1}} (q\pi)^1.$$

By Lemma 8 on the two fibres, this is equivalent to

$$F_q(\mu) \geq F_q(\nu) \iff F_{q\pi}(\mu\pi) \geq F_{q\pi}(\nu\pi).$$

The map $\mu \mapsto \mu\pi$ is an affine bijection from $\mathcal{M}_q$ to $\mathcal{M}_{q\pi}$, with inverse induced by π^{-1} . Therefore $G(\mu) := F_{q\pi}(\mu\pi)$ is a continuous affine representative of the same weak order on $\mathcal{M}_q$ as F_q . Herstein–Milnor uniqueness gives $G = aF_q + b$ with $a > 0$. Since $\delta_q\pi = \delta_{q\pi}$ and Lemma 9 gives

$$F_q(\delta_q) = F_{q\pi}(\delta_{q\pi}) = 0,$$

we have $b = 0$. Setting $c_{\pi,q} := a$ yields

$$F_{q\pi}(\mu\pi) = c_{\pi,q} F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$.

No-information equivalence. By Lemma 9, $F_q(\delta_q) = 0$ for every full-support q on every finite action set, independently of cardinality. $\square$

Lemma 11 (Undetectable distinctions neutrality). *Let $S : A \rightarrow A'$ be an onto deterministic partition map, and let $P : A \rightarrow \Delta(O)$ be S -measurable:*

$$S(a) = S(b) \implies P(\cdot | a) = P(\cdot | b).$$

Then for every full-support $q \in \Delta(A)$,

$$q^0 \sim_{P \sqcup S^q P} (qS)^1.$$

In particular, if $G_S : A \rightarrow \Delta(A')$ reveals the block $S(a)$, then

$$(q, G_S) \sim (qS, \text{Id}_{A'}).$$

Proof. Since S is onto and q has full support, qS has full support on A' . Thus the Bayesian pushforward $S^q P$ has no zero-row ambiguity in this lemma. View the deterministic map S as its associated stochastic kernel. One direction is exactly Axiom (A5):

$$q^0 \succeq_{P \sqcup S^q P} (qS)^1.$$

For the reverse direction, define the Bayesian refinement kernel $R : A' \rightarrow \Delta(A)$ by

$$R(a | a') = \begin{cases} q(a)/(qS)(a'), & S(a) = a', \\ 0, & \text{otherwise.} \end{cases}$$

For each $a' \in A'$, $R(\cdot | a')$ is stochastic because

$$\sum_{a \in A} R(a | a') = \frac{\sum_{a: S(a)=a'} q(a)}{(qS)(a')} = 1.$$

Moreover, for every $a \in A$,

$$((qS)R)(a) = \sum_{a' \in A'} (qS)(a')R(a | a') = (qS)(S(a)) \frac{q(a)}{(qS)(S(a))} = q(a) > 0.$$

Thus $(qS)R = q$.

Since P is S -measurable, the row of $S^q P$ at each block a' equals the common P -row on that block: if $S(a) = a'$, then

$$S^q P(o | a') = \frac{\sum_{b: S(b)=a'} q(b)P(o | b)}{(qS)(a')} = P(o | a).$$

Now pushing $S^q P$ back through R recovers P . For each $a \in A$,

$$(R^{qS}(S^q P))(o | a) = \frac{\sum_{a' \in A'} (qS)(a')R(a | a')S^q P(o | a')}{((qS)R)(a)} = S^q P(o | S(a)) = P(o | a).$$

Therefore

$$R^{qS}(S^q P) = P.$$

Apply Axiom (A5) to $(qS, S^q P)$ with the refinement kernel R :

$$(qS)^0 \succeq_{S^q P \sqcup P} q^1.$$

By Axiom (A3) applied to the two-block environment $P \sqcup S^q P$ with ordered pair $(1, 0)$, this is equivalent to

$$(qS)^1 \succeq_{P \sqcup S^q P} q^0.$$

Together the two inequalities give the desired indifference. $\square$

Lemma 12 (Support restriction). *Let $r \in \Delta(A)$ have support $B := \text{supp}(r) \subsetneq A$. Then:*

- (i) *Posterior-law reduction. For any channel $P : A \rightarrow \Delta(O)$, the posterior law $\mu_{r,P}$ depends only on the rows $P(\cdot | b)$ for $b \in B$. Rows $P(\cdot | a)$ with $a \notin B$ have zero contribution because $r(a) = 0$.*
- (ii) *Preference equivalence. Two channels $P, Q : A \rightarrow \Delta(O)$ that agree on B (i.e. $P(\cdot | b) = Q(\cdot | b)$ for all $b \in B$) are indifferent at prior r .*
- (iii) *Face identification. For every channel $P : A \rightarrow \Delta(O)$,*

$$(r, P) \sim (r|_B, P|_B)$$

as a two-block comparison; consequently the comparison between channels P, Q at prior r agrees with the comparison between $P|_B, Q|_B$ at the full-support prior $r|_B \in \Delta(B)$. This is proved by projecting A to B and embedding B back into A , using the all-completions form of Axiom (A5) for the embedding.

- (iv) *Representative. Define $F_r : \mathcal{M}_r \rightarrow \mathbb{R}$ by $F_r(\mu) := F_{r|_B}^B(\pi_B(\mu))$, where π_B projects posterior laws from $\Delta(A)$ to $\Delta(B)$ and $F_{r|_B}^B$ is the Lemma 8 representative on $\Delta(B)$. This F_r represents continuation comparisons at prior r in action set A .*

Proof. In this proof, $(s, R) \succeq (t, T)$ abbreviates the block comparison $s^0 \succeq_{R \sqcup T} t^1$, and $\sim$ denotes both weak inequalities. Whenever several such comparisons are chained, we first place the relevant blocks in a common finite block environment using Axiom (A3) and then apply Axiom (A1) transitivity.

Part (i): The marginal probability $m_{r,P}(o) = \sum_a r(a)P(o | a)$ and the Bayes numerator $r(a)P(o | a)$ are both zero for $a \notin B$. Hence $\mu_{r,P}$ is determined by $\{P(\cdot | b) : b \in B\}$.

Part (ii): Let P and $\tilde{P}$ be channels on A that agree on B . We first show $(r, P) \sim (r, \tilde{P})$. Fix $b_0 \in B$, and let $S : A \rightarrow B$ be the deterministic map with $S(b) = b$ for $b \in B$ and $S(a) = b_0$ for $a \notin B$. Then $rS = r|_B$ and the Bayesian pushforward $S^r P$ is $P|_B$ on B . Axiom (A5) gives

$$(r, P) \succeq (r|_B, P|_B).$$

Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$, and the Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on B and unconstrained on $A \setminus B$. By the all-completions convention in Axiom (A5), choose the completion to be $\tilde{P}$. Axiom (A5) gives

$$(r|_B, P|_B) \succeq (r, \tilde{P}).$$

Thus $(r, P) \succeq (r, \tilde{P})$. Applying the same projection/inclusion argument with $\tilde{P}$ as the original channel and choosing P as the inclusion completion gives $(r, \tilde{P}) \succeq (r, P)$. Hence $(r, P) \sim (r, \tilde{P})$.

If P, Q agree on B , choose a common extension $\tilde{P} = \tilde{Q}$ of this common restriction to all of A (for instance, use one fixed off-support row). The preceding paragraph gives $(r, P) \sim (r, \tilde{P})$ and $(r, Q) \sim (r, \tilde{Q})$; since $\tilde{P} = \tilde{Q}$, common-environment transitivity gives $(r, P) \sim (r, Q)$.

Part (iii): We first prove $(r, P) \sim (r|_B, P|_B)$ for every channel $P : A \rightarrow \Delta(O)$.

Forward direction (A5 projection). Let $S : A \rightarrow B$ map all $a \notin B$ to some fixed $b_0 \in B$. The coarsened prior is $rS = r|_B$ (since $r(a) = 0$ for $a \notin B$), and the Bayesian pushforward $S^r P$ equals $P|_B$ on B . By Axiom (A5),

$$(r, P) \succeq (r|_B, P|_B).$$

Reverse direction (A5 inclusion with completion). Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$. The Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on B and arbitrary on $A \setminus B$. Choose the Axiom (A5) completion to be any extension $\tilde{P} : A \rightarrow \Delta(O)$ of $P|_B$. Then

$$(r|_B, P|_B) \succeq (r, \tilde{P}).$$

By Part (ii), $(r, \tilde{P}) \sim (r, P)$ because $\tilde{P}$ and P agree on $B = \text{supp}(r)$. Hence $(r|_B, P|_B) \succeq (r, P)$.

Together the two directions give $(r, P) \sim (r|_B, P|_B)$.

Comparison equivalence. For any P, Q : $(r, P) \succeq (r, Q)$ iff (by the above indifferences $(r, P) \sim (r|_B, P|_B)$, $(r, Q) \sim (r|_B, Q|_B)$, and Axiom (A1) transitivity) $(r|_B, P|_B) \succeq (r|_B, Q|_B)$.

Part (iv): If $\mu \in \mathcal{M}_r$, every posterior in the finite support of μ is supported on B ; otherwise the barycentre r would assign positive mass to $A \setminus B$. Hence $\pi_B(\mu) \in \mathcal{M}_{r|_B}$ is well defined. By (iii), continuation comparisons at r are isomorphic to those at $r|_B$. The Lemma 8 representative $F_{r|_B}^B$ on $\Delta(B)$ therefore induces a representative on $\mathcal{M}_r$ via π_B . $\square$

Lemma 13 (Derived background inertness). *Assume Axioms (A1) and (A3)–(A6). Let A_1, A_2 be finite action sets and let $q_1 \in \Delta(A_1)$ and $q_2 \in \Delta(A_2)$ have full support. Then, for all channels P_1, Q_1 on A_1 and every background channel R_2 on A_2 ,*

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \iff q_1^0 \succeq_{P_1 \sqcup Q_1} q_1^1, \quad (\text{BI})$$

and symmetrically for comparisons in the second component with a common first-component background. In particular the left-hand comparison does not depend on the common background: for all backgrounds R_2, S_2 on A_2 ,

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \iff (q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes S_2) \sqcup (Q_1 \otimes S_2)} (q_1 \otimes q_2)^1.$$

Proof. Throughout, for a prior s on an action set A and channels V, W on A write $V \succeq_s W$ for the block comparison $s^0 \succeq_{V \sqcup W} s^1$, and $(s, V) \sim (t, W)$ for the two-block indifference between the pairs. By Lemma 3 these comparisons form a weak order for each fixed prior, and chains of comparisons between pairs are read in a common finite block environment using Axiom (A3) followed by Axiom (A1) transitivity. By Lemma 5, at a full-support prior a channel may be replaced by any channel with the same posterior law, in weak and in strict comparisons alike; we call this *substitution*.

Preliminary reduction: a common foreground alphabet. Axiom (A6) requires the two continuation channels used on a given branch to share an outcome alphabet. If $P_1 : A_1 \rightarrow \Delta(O)$ and $Q_1 : A_1 \rightarrow \Delta(O')$, replace them by their zero-padded versions on $O \sqcup O'$, as in Step 2 of Lemma 6. Padding adds only null outcomes, so it changes neither μ_{q_1} , nor $\mu_{q_1 \otimes q_2, \cdot \otimes R_2}$; both sides of (BI) are therefore unaffected by substitution at the full-support priors q_1 and $q_1 \otimes q_2$. Assume from now on that P_1 and Q_1 share the outcome alphabet O_1 .

For a channel $V : A_1 \rightarrow \Delta(O_1)$ and a finite set A_2 , write $\tilde{V}$ for its lift to the product action set,

$$\tilde{V}(z \mid (a_1, a_2)) := V(z \mid a_1),$$

and let $\tilde{R}_2(o_2 \mid (a_1, a_2)) := R_2(o_2 \mid a_2)$ be the lift of the background.

Step 1: a product channel is a two-stage channel with background first stage. Take $\tilde{R}_2$ as first stage on the action set $A_1 \times A_2$ and the constant continuation profile $\tilde{V}$ in every branch, so that every branch uses the same continuation alphabet O_1 and the compound outcome set $\bigsqcup_{o_2 \in O_2} (\{o_2\} \times O_1)$ is $O_2 \times O_1$. By the definition of two-stage composition,

$$(\tilde{R}_2 \triangleright \{\tilde{V}\}_{o_2})(o_2, z \mid (a_1, a_2)) = R_2(o_2 \mid a_2) V(z \mid a_1) = (V \otimes R_2)(z, o_2 \mid (a_1, a_2)).$$

The two channels therefore differ only by the bijection $(o_2, z) \mapsto (z, o_2)$ of outcome labels, which leaves every outcome probability and every posterior unchanged. Hence they induce the same posterior law at $q_1 \otimes q_2$, and substitution makes them interchangeable in every comparison at that prior. Applying this to $V = P_1$ and $V = Q_1$, the left-hand side of (BI) is equivalent to

$$\tilde{R}_2 \triangleright \{\tilde{P}_1\} \succeq_{q_1 \otimes q_2} \tilde{R}_2 \triangleright \{\tilde{Q}_1\}, \quad (*)$$

and likewise for the strict comparisons.

Step 2: every reached branch posterior is a product with first marginal q_1 . Write $q := q_1 \otimes q_2$, $m(o_2) := m_{q, \tilde{R}_2}(o_2)$ and, for $m(o_2) > 0$, $r_{o_2} := r_{o_2}^{q, \tilde{R}_2}$. Since the rows of $\tilde{R}_2$ do not depend on a_1 ,

$$m(o_2) = \sum_{a_2 \in A_2} q_2(a_2) R_2(o_2 \mid a_2) = m_{q_2, R_2}(o_2),$$

and Bayes' rule gives

$$r_{o_2}(a_1, a_2) = \frac{q_1(a_1) q_2(a_2) R_2(o_2 \mid a_2)}{m(o_2)} = q_1(a_1) q_2^{o_2}(a_2), \quad q_2^{o_2} := r_{o_2}^{q_2, R_2}.$$

Thus $r_{o_2} = q_1 \otimes q_2^{o_2}$: the background signal may change beliefs about a_2 , but in every reached branch the two components stay independent and the first marginal is still q_1 . At least one branch has positive probability.

Step 3: a lifted foreground channel is evaluated as on its own component. We claim that for every $s_2 \in \Delta(A_2)$ and every channel $V : A_1 \rightarrow \Delta(O_1)$,

$$(q_1 \otimes s_2, \tilde{V}) \sim (q_1, V). \quad (\dagger)$$

First suppose s_2 has full support, so that $q_1 \otimes s_2$ has full support on $A_1 \times A_2$. Let $S : A_1 \times A_2 \rightarrow A_1$ be the coordinate projection, an onto deterministic map, and note that $\tilde{V}$ is S -measurable, since $S(a) = S(b)$ implies $\tilde{V}(\cdot \mid a) = \tilde{V}(\cdot \mid b)$. Moreover $(q_1 \otimes s_2)S = q_1$ and, for every a_1 ,

$$(S^{q_1 \otimes s_2} \tilde{V})(z \mid a_1) = \frac{\sum_{a_2 \in A_2} q_1(a_1) s_2(a_2) V(z \mid a_1)}{q_1(a_1)} = V(z \mid a_1),$$

so the Bayesian pushforward of $\tilde{V}$ under S is V itself. Undetectable-distinctions neutrality (Lemma 11) gives $(\dagger)$.

If s_2 is not full support, let $B_2 := \text{supp}(s_2)$, so that $B := A_1 \times B_2$ is the support of $q_1 \otimes s_2$ and $B \subsetneq A_1 \times A_2$. The set B is the product action set $A_1 \times B_2$, so no relabelling is involved: the restricted prior is $q_1 \otimes (s_2|_{B_2})$ and the restricted channel $\tilde{V}|_B$ is the lift of V to $A_1 \times B_2$. Face identification (Lemma 12(iii)) gives $(q_1 \otimes s_2, \tilde{V}) \sim (q_1 \otimes s_2|_{B_2}, \tilde{V}|_B)$, and the full-support case just proved, applied on the action set $A_1 \times B_2$, gives $(q_1 \otimes s_2|_{B_2}, \tilde{V}|_B) \sim (q_1, V)$. Transitivity in a common block environment yields $(\dagger)$ in general.

Fix a branch with $m(o_2) > 0$ and apply $(\dagger)$ at $s_2 = q_2^{o_2}$ to $V = P_1$ and to $V = Q_1$, using $r_{o_2} = q_1 \otimes q_2^{o_2}$ from Step 2. Place the four pairs (r_{o_2}, P_1) , (q_1, P_1) , (r_{o_2}, Q_1) , (q_1, Q_1) as blocks of one finite block environment (Axiom (A3), Lemma 3). The two indifferences and transitivity of the weak order in that environment (Axiom (A1)) give

$$\tilde{P}_1 \succeq_{r_{o_2}} \tilde{Q}_1 \iff P_1 \succeq_{q_1} Q_1, \quad (\dagger)$$

and, since the relation is complete, the same equivalence for the strict comparisons. In particular the branch comparison is the same in every reached branch and does not involve R_2 .

Step 4: forward implication. Assume $P_1 \succeq_{q_1} Q_1$. By $(\dagger)$, the branch hypothesis $r_{o_2}^0 \succeq_{\tilde{P}_1 \sqcup \tilde{Q}_1} r_{o_2}^1$ of Axiom (A6) holds for every branch with $m(o_2) > 0$, with the common first stage $\tilde{R}_2$ and the two constant continuation profiles. Axiom (A6) yields $(*)$, hence the left-hand side of (BI) by Step 1.

Step 5: converse, via the strictness clause. Assume the left-hand side of (BI), so that $(*)$ holds by Step 1, and suppose $P_1 \succeq_{q_1} Q_1$ fails. By completeness and the reverse-block equivalence of Lemma 3, $Q_1 \succ_{q_1} P_1$, so by the strict form of $(\dagger)$ every reached branch satisfies $\tilde{Q}_1 \succ_{r_{o_2}} \tilde{P}_1$. At least one such branch exists and has positive probability, so the strictness clause of Axiom (A6), applied with the two profiles exchanged, gives

$$\tilde{R}_2 \triangleright \{\tilde{Q}_1\} \succ_{q_1 \otimes q_2} \tilde{R}_2 \triangleright \{\tilde{P}_1\},$$

contradicting $(*)$ because $\succeq_{q_1 \otimes q_2}$ is a weak order (Lemma 3). Hence $P_1 \succeq_{q_1} Q_1$, proving (BI).

The second-component clause is the same argument with the roles of the two coordinates exchanged: the common first-component background is used as the first stage and the second-component channels as the continuations. Finally, since both product comparisons in the last display of the lemma are equivalent to the single background-free comparison $P_1 \succeq_{q_1} Q_1$, they are equivalent to each other. $\square$

The lemma shows that separability across independent components is not an independent behavioural commitment. A product channel is a two-stage channel whose first stage is the background experiment: it can teach the observer about the background action, but it leaves the foreground component independent of it and its marginal unchanged, so in every reached branch the continuation comparison is the foreground comparison itself. Branchwise monotonicity then transmits that single comparison to the compound, and its strictness clause transmits it back. The derived statement is stronger than mere invariance to changing the background: it identifies the product comparison with the comparison of the foreground channels alone, which is the form used in Lemma 14 below.

Lemma 14 (Coherent product quasi-additivity). *There is a choice of positive rescalings of the zero-normalised fibrewise representatives F_q , still denoted F_q , and a scalar $\kappa \in \mathbb{R}$ such that, for all full-support priors $p \in \Delta(A)$ and $r \in \Delta(B)$ and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$,*

$$F_{p \otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) + \kappa F_p(\mu) F_r(\nu). \quad (\text{QA})$$

If one factor is a singleton, the same formula holds with its identically zero representative. Moreover, for every nondegenerate r and every $\nu \in \mathcal{M}_r$,

$$1 + \kappa F_r(\nu) > 0. \quad (\text{POS})$$

Proof. All representatives in this proof are zero-normalised: $F_q(\delta_q) = 0$. The proof first treats nondegenerate action sets, meaning action sets with at least two elements.

Step 1: product-coordinate independence. Fix full-support priors p and r on nondegenerate finite action sets and set

$$L_{p,r}(\mu, \nu) := F_{p \otimes r}(\mu \otimes \nu).$$

This map is continuous and separately affine. To identify a first-coordinate slice, let P, Q implement $\mu, \mu' \in \mathcal{M}_p$ and let R implement $\nu \in \mathcal{M}_r$. By the product factorisation (PL), $P \otimes R$ implements $\mu \otimes \nu$ and $Q \otimes R$ implements $\mu' \otimes \nu$ at the full-support prior $p \otimes r$. Hence Lemma 8, applied on the fibre $\mathcal{M}_{p \otimes r}$ and on the fibre $\mathcal{M}_p$, turns background inertness (BI) of Lemma 13 into

$$L_{p,r}(\mu, \nu) \geq L_{p,r}(\mu', \nu) \iff F_p(\mu) \geq F_p(\mu').$$

The symmetric clause of Lemma 13 identifies every second-coordinate slice with the F_r -order. Local non-triviality, the fibrewise representation, and the zero normalisation from Lemma 9 make both slice orders nonconstant: $F_p(\chi_p) > 0$ and $F_r(\chi_r) > 0$.

Step 2: the pair-specific bilinear form. Put $x(\mu) := F_p(\mu)$ and $y(\nu) := F_r(\nu)$. For fixed ν , affine-utility uniqueness gives

$$L_{p,r}(\mu, \nu) = \alpha(\nu)x(\mu) + \gamma(\nu), \quad \alpha(\nu) > 0.$$

Taking $\mu = \delta_p$ shows $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$. This is a continuous affine representative of the F_r -order and vanishes at δ_r , so

$$\gamma(\nu) = B_{p,r}y(\nu)$$

for some $B_{p,r} > 0$. Likewise, the slice at $\nu = \delta_r$ gives

$$L_{p,r}(\mu, \delta_r) = A_{p,r}x(\mu)$$

for some $A_{p,r} > 0$.

Let $\bar{\mu} := \chi_p$ and $h_p := F_p(\chi_p) > 0$. The function $\nu \mapsto L_{p,r}(\bar{\mu}, \nu)$ is again a continuous affine representative of the F_r -order. Hence there are $D_{p,r} > 0$ and a constant $E_{p,r}$ such that

$$L_{p,r}(\bar{\mu}, \nu) = D_{p,r}y(\nu) + E_{p,r}.$$

At $\nu = \delta_r$, $E_{p,r} = A_{p,r}h_p$. Comparing this identity with $L_{p,r}(\bar{\mu}, \nu) = \alpha(\nu)h_p + B_{p,r}y(\nu)$ gives

$$\alpha(\nu) = A_{p,r} + C_{p,r}y(\nu), \quad C_{p,r} := \frac{D_{p,r} - B_{p,r}}{h_p}.$$

Therefore

$$L_{p,r}(\mu, \nu) = A_{p,r}x(\mu) + B_{p,r}y(\nu) + C_{p,r}x(\mu)y(\nu). \quad (\text{BIL})$$

The coordinate-slice slopes are positive:

$$A_{p,r} + C_{p,r}y(\nu) > 0, \quad B_{p,r} + C_{p,r}x(\mu) > 0.$$

Step 3: normalising the two linear coefficients. Adopt the strict-product convention: finite products are identified by tuple notation, so $(A \times B) \times C = A \times B \times C = A \times (B \times C)$ and similarly for product priors. This convention does not identify $A \times B$ with $B \times A$.

Expanding $F_{p \otimes r \otimes s}$ under the two groupings and comparing the coefficients of x , y , and z gives

$$A_{p \otimes r, s} A_{p, r} = A_{p, r \otimes s}, \quad (\text{C1})$$

$$A_{p \otimes r, s} B_{p, r} = B_{p, r \otimes s} A_{r, s}, \quad (\text{C2})$$

$$B_{p \otimes r, s} = B_{p, r \otimes s} B_{r, s}. \quad (\text{C3})$$

Coefficient comparison is legitimate because the mixtures $(1-t)\delta_p + t\chi_p$ make x range over a nondegenerate interval, and similarly in the other coordinates.

The coordinate swap gives a scalar $d(p, r) > 0$ such that

$$F_{r \otimes p}(\nu \otimes \mu) = d(p, r) F_{p \otimes r}(\mu \otimes \nu).$$

Comparing coefficients in (BIL),

$$B_{r,p} = d(p, r) A_{p,r}, \quad A_{r,p} = d(p, r) B_{p,r}, \quad C_{r,p} = d(p, r) C_{p,r}.$$

Thus, with $\rho(p, r) := B_{p,r}/A_{p,r}$,

$$\rho(r, p) = \rho(p, r)^{-1}. \quad (\text{SW})$$

Divide (C2) by (C1):

$$\rho(p, r) = \rho(p, r \otimes s) A_{r,s}.$$

Fix a nondegenerate reference prior q_0 on a finite action set A_0 and put $\phi(r) := \rho(q_0, r)$. Then

$$A_{r,s} = \frac{\phi(r)}{\phi(r \otimes s)}. \quad (\text{G})$$

Rescale every representative by $F_p^* := \phi(p) F_p$. Under this positive rescaling,

$$A_{p,r}^* = A_{p,r} \frac{\phi(p \otimes r)}{\phi(p)}, \quad B_{p,r}^* = B_{p,r} \frac{\phi(p \otimes r)}{\phi(r)}, \quad C_{p,r}^* = C_{p,r} \frac{\phi(p \otimes r)}{\phi(p)\phi(r)}.$$

Equation (G) gives $A_{p,r}^* = 1$ for every pair. Write $\beta(p, r) := B_{p,r}^*$. The transformed versions of (C2)–(C3) and (SW) are

$$\begin{aligned} \beta(p, r) &= \beta(p, r \otimes s), \\ \beta(p \otimes r, s) &= \beta(p, r) \beta(r, s), \\ \beta(r, p) &= \beta(p, r)^{-1}. \end{aligned}$$

Moreover $\phi(q_0) = \rho(q_0, q_0) = 1$. Equation (G) gives $A_{q_0,p} = 1/\phi(q_0 \otimes p)$, while $B_{q_0,p} = A_{q_0,p} \rho(q_0, p) = \phi(p)/\phi(q_0 \otimes p)$. Therefore

$$\beta(q_0, p) = B_{q_0,p} \frac{\phi(q_0 \otimes p)}{\phi(p)} = 1.$$

By swap reciprocity, $\beta(p, q_0) = 1$. Taking $s = q_0$ in the second displayed identity gives

$$1 = \beta(p \otimes r, q_0) = \beta(p, r) \beta(r, q_0) = \beta(p, r).$$

Thus, after the positive gauge rescaling, $A_{p,r} = B_{p,r} = 1$ for every nondegenerate pair. Positivity of the slice slopes is preserved because the rescaling factors are positive.

Step 4: the interaction coefficient is universal. Write the remaining coefficient as $\kappa_{p,r}$. Comparing the xy , xz , yz , and xyz coefficients in the two associative expansions gives

$$\kappa_{p,r} = \kappa_{p,r \otimes s}, \quad (\text{K1})$$

$$\kappa_{p \otimes r, s} = \kappa_{p, r \otimes s}, \quad (\text{K2})$$

$$\kappa_{p \otimes r, s} = \kappa_{r, s}, \quad (\text{K3})$$

$$\kappa_{p \otimes r, s} \kappa_{p, r} = \kappa_{p, r \otimes s} \kappa_{r, s}. \quad (\text{K4})$$

The last equation follows from the first three, but it records the full coefficient comparison. After $A = B = 1$, the linear coefficient comparison under the coordinate swap forces the swap scalar to be one; the bilinear coefficient comparison then gives

$$\kappa_{p,r} = \kappa_{r,p}. \quad (\text{KS})$$

Using (K1)–(K3) with the reference prior q_0 ,

$$\kappa_{p,r} = \kappa_{p,r \otimes q_0} = \kappa_{p \otimes r, q_0} = \kappa_{r, q_0}.$$

Hence $\kappa_{p,r}$ is independent of its first argument. By symmetry it is also independent of its second argument. Denote the common value by κ . This proves (QA) for nondegenerate factors.

The positive slice-slope condition now reads

$$1 + \kappa F_r(\nu) > 0,$$

which is (POS).

Step 5: singleton factors. Let $q_* = \delta_*$ be the prior on a singleton. Its zero-normalised representative is identically zero. If p is nondegenerate, the canonical bijection $A \rightarrow A \times \{*\}$ and Lemma 10 give a scalar $c > 0$ with

$$F_{p \otimes q_*}(\mu \otimes \delta_{q_*}) = c F_p(\mu).$$

Tensor both sides with the nondegenerate reference prior q_0 . The canonical bijection between $(A \times \{*\}) \times A_0$ and $A \times A_0$ gives a scalar $d > 0$, and quasi-additivity for the two nondegenerate products gives, with $x = F_p(\mu)$ and $y = F_{q_0}(\nu)$,

$$cx + y + \kappa cxy = d(x + y + \kappa xy).$$

Taking $\mu = \delta_p$ and $\nu = \chi_{q_0}$ gives $x = 0$ and $y > 0$, hence $d = 1$. Taking $\mu = \chi_p$ and $\nu = \delta_{q_0}$ gives $x > 0$ and $y = 0$, hence $c = 1$. Thus (QA) also holds with a singleton factor. The reversed order is identical. $\square$

Corollary 15 (Exact relabelling invariance). *For every bijection $\pi : A \rightarrow A'$ and every full-support $q \in \Delta(A)$,*

$$F_{q\pi}(\mu\pi) = F_q(\mu) \quad (\mu \in \mathcal{M}_q).$$

Proof. The singleton case is immediate because both representatives vanish. Suppose A is nondegenerate. Lemma 10 gives $F_{q\pi}(\mu\pi) = c F_q(\mu)$ for some $c > 0$. Apply the product bijection $\pi \times \text{id}_{A_0}$ with the nondegenerate reference prior q_0 . There is $d > 0$ such that, writing $x = F_q(\mu)$ and $y = F_{q_0}(\nu)$,

$$cx + y + \kappa cxy = d(x + y + \kappa xy).$$

Set $\mu = \delta_q$ and $\nu = \chi_{q_0}$, so $x = 0$ and $y > 0$, to obtain $d = 1$. Then set $\mu = \chi_q$ and $\nu = \delta_{q_0}$, so $x > 0$ and $y = 0$, to obtain $c = 1$. $\square$

Lemma 16 (Cross-prior block representation). *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels $P : A \rightarrow \Delta(O)$ and $Q : B \rightarrow \Delta(Y)$,*

$$q^0 \succeq_{P \sqcup Q} r^1 \iff F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}). \quad (\text{BC})$$

Boundary representatives are interpreted by Lemma 12.

Proof. First suppose q and r have full support, allowing singleton action sets. At prior $q \otimes r$, quasi-additivity and zero normalisation give

$$F_{q \otimes r}(\mu_{q,P} \otimes \delta_r) = F_q(\mu_{q,P}), \quad F_{q \otimes r}(\delta_q \otimes \mu_{r,Q}) = F_r(\mu_{r,Q}).$$

These values are in the same fibre $\mathcal{M}_{q \otimes r}$, so Lemma 8 gives

$$(q \otimes r)^0 \succeq_{(P \otimes U_B) \sqcup (U_A \otimes Q)} (q \otimes r)^1 \iff F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}).$$

It remains to transfer the product-lifted comparison back to the original two blocks. For the A -side pair, Axiom (A5) applied to the projection $\pi_A : A \times B \rightarrow A$ gives

$$(q \otimes r, P \otimes U_B) \succeq (q, P)$$

in block comparisons. Conversely, let $T : A \rightarrow \Delta(A \times B)$ be

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b).$$

Then $qT = q \otimes r$, and the Bayesian pushforward $T^q P$ has row $P(\cdot \mid a)$ at every (a, b) . Thus $T^q P$ is $P \otimes U_B$ after the canonical one-outcome identification of the B -experiment, and Axiom (A5) gives the reverse comparison. Equivalently, if the harmless outcome identification is not made, Lemma 5 identifies $T^q P$ and $P \otimes U_B$ because they induce the same posterior law at $q \otimes r$. Hence $(q \otimes r, P \otimes U_B)$ and (q, P) are indifferent in every relevant block comparison. The same projection/embedding argument identifies $(q \otimes r, U_A \otimes Q)$ with (r, Q) .

Place the four pairs (q, P) , $(q \otimes r, P \otimes U_B)$, (r, Q) , and $(q \otimes r, U_A \otimes Q)$ as blocks of one finite block environment. Lemma 3 and transitivity transfer the two pairwise indifferences to the two-block comparison, proving (BC) for full-support priors.

For arbitrary priors, let $A_+ := \text{supp}(q)$ and $B_+ := \text{supp}(r)$. Lemma 12 identifies (q, P) with $(q|_{A_+}, P|_{A_+})$, identifies (r, Q) with $(r|_{B_+}, Q|_{B_+})$, and identifies the two values $F_q(\mu_{q,P})$ and $F_r(\mu_{r,Q})$ with their support-face values. Applying the full-support bridge on A_+ and B_+ and pulling back by Lemma 12 proves the claim. Singleton supports use the same zero-representative convention as Lemma 14. $\square$

Lemma 17 (Branch aggregation). *Let $P_1 : A \rightarrow \Delta(O_1)$ be a first-stage channel at full-support q , with branch probabilities $m(o)$ and posteriors r_o . There are positive branch coefficients $\beta(q, r_o)$, depending only on q and the reached posterior r_o , such that for every continuation profile $\{Q^o\}_{o:m(o)>0}$,*

$$F_q(\mu_{q,P_1 \triangleright \{Q^o\}}) = F_q(\mu_{q,P_1}) + \sum_{o:m(o)>0} m(o) \beta(q, r_o) F_{r_o}(\mu_{r_o, Q^o}).$$

If r_o has full support, $\beta(q, r_o)$ is the full-support tangent coefficient. If $1 < |\text{supp}(r_o)| < |A|$, it is the full-to-face coefficient. If $|\text{supp}(r_o)| = 1$, the value F_{r_o} is identically zero and $\beta(q, r_o)$ is an arbitrary positive convention.

Proof. Fix the first-stage channel P_1 and its branch structure $(m(o), r_o)_{o:m(o)>0}$. For any profile of continuation channels $\{Q^o\}$, the compound posterior law at prior q is

$$\mu_{q,P_1 \triangleright \{Q^o\}} = \sum_{o:m(o)>0} m(o) \mu_{r_o, Q^o}.$$

Setup: linear part of affine functional. Since F_q is affine on $\mathcal{M}_q$ (Lemma 8), it extends uniquely to an affine functional on the affine hull of $\mathcal{M}_q$. Write L_q for the linear part: for any $\mu, \mu' \in \mathcal{M}_q$,

$$F_q(\mu) - F_q(\mu') = L_q(\mu - \mu').$$

Branch-affine structure. Fix a positive-probability branch $\bar{o}$ and fix continuation channels $H^o : A \rightarrow \Delta(Z^o)$ in all other branches $o \neq \bar{o}$. Let $r := r_{\bar{o}}$ and $m := m(\bar{o})$. If $\text{supp}(r)$ is a singleton, then $\mathcal{M}_r = \{\delta_r\}$ and $F_r(\delta_r) = 0$ by the singleton zero-normalisation convention; such branches contribute zero and their coefficient convention is recorded in Case 1 below. For the branch-affine argument in this paragraph, assume $|\text{supp}(r)| \geq 2$. If r is a boundary posterior, F_r and continuation comparisons at r are understood through Lemma 12.

Let

$$\rho_H := \sum_{o \neq \bar{o}} m(o) \mu_{r_o, H^o}.$$

For any $\nu \in \mathcal{M}_r$, the aggregate posterior law produced by using branch law ν in branch $\bar{o}$ and the fixed continuations elsewhere is

$$T_H(\nu) := m\nu + \rho_H \in \mathcal{M}_q.$$

Define $g : \mathcal{M}_r \rightarrow \mathbb{R}$ by

$$g(\nu) := F_q(T_H(\nu)).$$

Thus g depends only on ν , and $g = F_q \circ T_H$ is continuous and affine.

We now show that g and F_r represent the same weak order on $\mathcal{M}_r$. For $\nu, \nu' \in \mathcal{M}_r$, choose finite continuation channels implementing them at prior r ; when r is boundary these are implemented on $B = \text{supp}(r)$ and extended arbitrarily to A , as licensed by Lemma 12. Padding outcome alphabets if needed, Axiom (A6) compares the two continuation profiles that differ only in branch $\bar{o}$. If $F_r(\nu) > F_r(\nu')$, Axiom (A6) gives $g(\nu) > g(\nu')$. Conversely, if $g(\nu) \geq g(\nu')$ but $F_r(\nu) < F_r(\nu')$, applying the strict part of Axiom (A6) to the swapped branch comparison gives $g(\nu') > g(\nu)$, a contradiction. Hence $g(\nu) \geq g(\nu')$ iff $F_r(\nu) \geq F_r(\nu')$.

Since $|\text{supp}(r)| \geq 2$, local non-triviality on the support face and Lemmas 12 and 9 imply F_r is nonconstant. The continuous affine functions g and F_r are therefore nonconstant affine representatives of the same weak order on $\mathcal{M}_r$. Herstein–Milnor uniqueness gives

$$g(\nu) = \alpha F_r(\nu) + \gamma$$

for some $\alpha > 0$ and $\gamma \in \mathbb{R}$.

The coefficient α is independent of the fixed continuations. Write $g^{(1)}$ and $g^{(2)}$ for the functionals arising from two different choices $\{H^o\}$ and $\{\tilde{H}^o\}$ in the other branches. Both represent the same order as F_r , so $g^{(1)} = \alpha_1 F_r + \gamma_1$ and $g^{(2)} = \alpha_2 F_r + \gamma_2$ with $\alpha_1, \alpha_2 > 0$. To show $\alpha_1 = \alpha_2$, consider the difference $g^{(1)} - g^{(2)} = (\alpha_1 - \alpha_2)F_r + (\gamma_1 - \gamma_2)$. For any $\nu \in \mathcal{M}_r$,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = F_q\left(m(\bar{o})\nu + \sum_{o \neq \bar{o}} m(o)\mu_{r_o, H^o}\right) - F_q\left(m(\bar{o})\nu + \sum_{o \neq \bar{o}} m(o)\mu_{r_o, \tilde{H}^o}\right).$$

Both arguments are posterior laws in $\mathcal{M}_q$, differing only in the fixed non- $\bar{o}$ components. Since L_q has already been defined on the affine-hull difference space,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = L_q(\rho_1 - \rho_2),$$

where ρ_i is the fixed non- $\bar{o}$ background law for $g^{(i)}$. This difference is independent of ν . Since F_r is nonconstant in the present nondegenerate case, $\alpha_1 = \alpha_2$.

Path-independence of the branch coefficient via tangent-space decomposition. We now prove that $\alpha/m(\bar{o})$ depends only on the pair $(q, r_{\bar{o}})$, not on the first-stage experiment. This is the key step that converts ordinal branchwise monotonicity into a cardinal aggregation formula.

Tangent space at a posterior. For a posterior $r \in \Delta(A)$ with support $B := \text{supp}(r)$, define the *tangent space*:

$$V_r := \text{span}\{\nu - \nu' : \nu, \nu' \in \mathcal{M}_r\}.$$

Elements of V_r are signed measures with total mass zero and barycentre zero.

Tangent-space identification. For any r with full support, we claim

$$V_r = W := \{\eta \in \text{span } \Delta_f(\Delta(A)) : \eta(\Delta(A)) = 0, \int p d\eta(p) = 0\},$$

the space of finite-support signed measures on $\Delta(A)$ with zero total mass and zero barycentre. Indeed, if $\nu, \nu' \in \mathcal{M}_r$, then $\nu - \nu'$ has total mass $1 - 1 = 0$ and barycentre $r - r = 0$, so $V_r \subset W$. Conversely, given any $\eta \in W$: if $\eta = 0$ then $\eta \in V_r$ trivially. Otherwise, since η has finite support, write its Jordan decomposition $\eta = \eta^+ - \eta^-$ where $\eta^+, \eta^- \geq 0$ are finite-support positive measures with equal total mass $c > 0$. Normalise $\zeta^+ := \eta^+/c$ and $\zeta^- := \eta^-/c$; both are finite-support

probability measures on $\Delta(A)$ with the same barycentre $b := \int p d\zeta^+(p) = \int p d\zeta^-(p)$. Since $b \in \Delta(A)$, the set $\{a : b(a) > 0\}$ is nonempty. Because r has full support,

$$\theta := \min\left\{1, \min_{a: b(a) > 0} \frac{r(a)}{b(a)}\right\} > 0.$$

Choose $0 < \lambda < \theta$ and set

$$\tau := \frac{r - \lambda b}{1 - \lambda}.$$

Then $1 - \lambda > 0$, $\sum_a \tau(a) = 1$, and $\tau(a) \geq 0$ for every a : if $b(a) > 0$ this follows from $\lambda < r(a)/b(a)$, while if $b(a) = 0$ the numerator is $r(a) > 0$. Hence $\tau \in \Delta(A)$. Define

$$\nu := \lambda \zeta^+ + (1 - \lambda) \delta_\tau, \quad \nu' := \lambda \zeta^- + (1 - \lambda) \delta_\tau.$$

Both are finite-support probability measures on $\Delta(A)$ with barycentre $\lambda b + (1 - \lambda) \tau = r$, so $\nu, \nu' \in \mathcal{M}_r$. Then $\nu - \nu' = \lambda(\zeta^+ - \zeta^-) = (\lambda/c)\eta$, hence $\eta = (c/\lambda)(\nu - \nu') \in V_r$. Thus $W \subset V_r$ and $V_r = W$. Since W depends only on the full-support property (not on the specific prior), $V_q = V_r = V_s = W$ for all full-support priors.

Case 1: Degenerate branch posterior ($|B| = 1$). If $r = \delta_a$ for some action a , then $\mathcal{M}_{\delta_a} = \{\delta_{\delta_a}\}$ is a singleton: the only Bayes-plausible law with barycentre δ_a is the point mass at δ_a itself. Hence $V_r = \{0\}$, and every continuation experiment at prior δ_a gives the same posterior law δ_{δ_a} . We set $F_{\delta_a}(\delta_{\delta_a}) = 0$. For such branches, the term $m(o)\beta(q, r_o)F_{r_o}(\mu_{r_o, Q^o})$ is zero under any convention for $\beta(q, r_o)$.

For the aggregation formula below, assign any positive value to $\beta(q, \delta_a)$; the choice is not behaviourally identified because the corresponding continuation value is always zero.

Case 2: Non-full-support branch posterior ($1 < |B| < |A|$). If r has support $B \subsetneq A$ with $|B| \geq 2$, apply Lemma 12. Continuation comparisons at prior r reduce to those at $r|_B \in \Delta(B)$, with representative $F_r := F_{r|_B}^B \circ \pi_B$.

For any finite action set C , write

$$W_C := \{\sigma \in \text{span } \Delta_f(\Delta(C)) : \sigma(\Delta(C)) = 0, \int_{\Delta(C)} p d\sigma(p) = 0\}.$$

Let $j_B : \Delta(B) \rightarrow \Delta(A)$ be the face inclusion, extending each posterior by zero outside B , and let $i_B := (j_B)_\#$ act on finite signed posterior laws. By Lemma 12(iv), $F_r^A = F_{r|_B}^B \circ \pi_B$, so on face directions

$$L_r^A(i_B \sigma) = L_{r|_B}^B(\sigma) \quad (\sigma \in W_B).$$

Since $|B| \geq 2$ and $r|_B$ is full support on B , local non-triviality and Lemma 9 give

$$L_{r|_B}^B(\chi_{r|_B} - \delta_{r|_B}) > 0,$$

so $L_{r|_B}^B \neq 0$.

Full-to-face scalar. Fix any full-support $q \in \Delta(A)$. The boundary posterior r is reachable from q : choose

$$0 < \varepsilon < \min\left\{1, \min_{b \in B} \frac{q(b)}{r(b)}\right\}$$

and set $P_1(1 | a) = \varepsilon r(a)/q(a)$, with $r(a) = 0$ for $a \notin B$. Then branch 1 has positive probability ε and posterior r . For any nonzero $\sigma \in W_B$, the tangent-space identification on the face B gives $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}_{r|_B}^B$. Implement ν, ν' by continuation channels on B and extend them arbitrarily to A . Lemma 12 identifies their comparison at the boundary prior r with their comparison at $r|_B$.

Axiom (A6) applied to the branch reaching r , with all other branches fixed, gives the positive sign direction for feasible differences. The negative direction follows by swapping the two continuation laws, and equality follows by applying the weak part of Axiom (A6) in both

directions. Extending by positive scalar multiples from feasible differences to all of W_B , $L_q^A \circ i_B$ and $L_{r|B}^B$ have the same sign partition on W_B . Since $L_{r|B}^B \neq 0$, both are nonzero and there is a unique $\beta_A(q, r) > 0$ such that

$$L_q^A(i_B \sigma) = \beta_A(q, r) L_{r|B}^B(\sigma) \quad (\sigma \in W_B).$$

This is the ambient full-to-face coefficient used for boundary branches.

Case 3: Full-support branch posterior ($B = A$). This is the main case. Suppose r has full support on A .

Claim 1: aggregate change from branch variation. If a first-stage branch reaches posterior r with probability m , and the branch continuation law changes from ν to ν' (both in $\mathcal{M}_r$), then the aggregate posterior law changes by $m(\nu - \nu') \in m \cdot V_r$. This difference depends only on (m, ν, ν', r) —not on which experiment produced the branch, nor on the other branches.

Proof of Claim 1. The aggregate changes from $m\nu + \rho$ to $m\nu' + \rho$, where ρ collects the other branches. The difference is $m(\nu - \nu')$.

Claim 2: $L_q|_{V_r}$ is a positive scalar multiple of $L_r|_{V_r}$.

Proof of Claim 2. We first show that every direction in V_r is proportional to a feasible difference. By the Tangent-space identification above, any nonzero $\sigma \in V_r$ can be written as $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}_r$. Hence every nonzero $\sigma \in V_r$ is a positive scalar multiple of some feasible difference.

By Axiom (A6) (branchwise monotonicity), if $\nu \succ_r \nu'$ (i.e. $F_r(\nu) > F_r(\nu')$, equivalently $L_r(\nu - \nu') > 0$), then the aggregate comparison ranks the ν -continuation strictly higher. Since the aggregate value difference is $L_q(m(\nu - \nu')) = mL_q(\nu - \nu')$ with $m > 0$, we have $L_q(\nu - \nu') > 0$ whenever $L_r(\nu - \nu') > 0$.

Reverse and equality directions. If $L_r(\nu - \nu') < 0$, completeness of $\succeq_r$ gives $\nu' \succ_r \nu$, and A6 applied to the reversed comparison gives $L_q(\nu' - \nu) > 0$, i.e. $L_q(\nu - \nu') < 0$. If $L_r(\nu - \nu') = 0$, then $\nu \sim_r \nu'$, so both $\nu \succeq_r \nu'$ and $\nu' \succeq_r \nu$; applying A6 weakly in both directions gives $L_q(\nu - \nu') \geq 0$ and $L_q(\nu - \nu') \leq 0$, hence $L_q(\nu - \nu') = 0$.

Since every nonzero $\sigma \in V_r$ is proportional to some feasible difference, the sign agreement extends from feasible differences to all of V_r : for any $\sigma \in V_r$, $L_q(\sigma) > 0$ iff $L_r(\sigma) > 0$, $L_q(\sigma) < 0$ iff $L_r(\sigma) < 0$, and $L_q(\sigma) = 0$ iff $L_r(\sigma) = 0$.

Two nonzero linear functionals on a vector space that have the same positive half-space must be positive scalar multiples of each other. (If $L_r \neq 0$, then $\ker L_q = \ker L_r$ and they lie in the same half-space, so $L_q = cL_r$ for some $c > 0$.) Since r has full support and local non-triviality gives $F_r(\chi_r) > F_r(\delta_r) = 0$, we have $L_r \neq 0$. Hence there exists $\beta(q, r) > 0$ such that

$$L_q(\sigma) = \beta(q, r) L_r(\sigma) \quad \text{for all } \sigma \in V_r.$$

Path-independence. Consider any first-stage experiment reaching posterior r in some branch with probability m . Changing the branch continuation from ν to ν' changes the aggregate value by

$$L_q(m(\nu - \nu')) = m \beta(q, r) L_r(\nu - \nu') = m \beta(q, r) (F_r(\nu) - F_r(\nu')).$$

The coefficient $m \beta(q, r)$ depends only on (q, r, m) . In particular, $\beta(q, r)$ is independent of which experiment reached r and independent of the other branches.

Comparing with the earlier notation $g(\nu) = \alpha F_r(\nu) + \gamma$: the value difference is $g(\nu) - g(\nu') = \alpha(F_r(\nu) - F_r(\nu'))$, so $\alpha = m \beta(q, r)$ and hence $\alpha/m = \beta(q, r)$ universally. The coefficient $\beta(q, r_{\bar{o}})$ is defined as the scalar from $L_q|_{V_{r_{\bar{o}}}} = \beta(q, r_{\bar{o}}) L_{r_{\bar{o}}}|_{V_{r_{\bar{o}}}}$; it depends only on q and $r_{\bar{o}}$, not on the experiment that produced the branch.

Aggregation formula. Let

$$\mu^0 := \sum_{o: m(o) > 0} m(o) \delta_{r_o} = \mu_{q, P_1}.$$

For each positive-probability branch let $B_o := \text{supp}(r_o)$. If $B_o = A$, set $v_o := \mu_{r_o, Q^o} - \delta_{r_o} \in W_A$. If $1 < |B_o| < |A|$, set

$$\bar{v}_o := \pi_{B_o}(\mu_{r_o, Q^o}) - \delta_{r_o|_{B_o}} \in W_{B_o}, \quad v_o := i_{B_o} \bar{v}_o \in W_A.$$

If $|B_o| = 1$, set $v_o := 0$. Then

$$\mu_{q, P_1 \triangleright \{Q^o\}} = \mu^0 + \sum_{o: m(o) > 0} m(o) v_o.$$

Since $V_q = W_A$, affinity of F_q gives

$$F_q(\mu_{q, P_1 \triangleright \{Q^o\}}) - F_q(\mu^0) = \sum_{o: m(o) > 0} m(o) L_q^A(v_o).$$

For a full-support branch $B_o = A$, Case 3 gives

$$L_q^A(v_o) = \beta(q, r_o) L_{r_o}^A(v_o) = \beta(q, r_o) F_{r_o}(\mu_{r_o, Q^o}).$$

For a nondegenerate boundary branch $1 < |B_o| < |A|$, Case 2 gives

$$L_q^A(v_o) = L_q^A(i_{B_o} \bar{v}_o) = \beta_A(q, r_o) L_{r_o|_{B_o}}^{B_o}(\bar{v}_o) = \beta_A(q, r_o) F_{r_o}(\mu_{r_o, Q^o}).$$

The last equalities use zero normalisation: Lemma 9 in full support, Lemma 12 plus Lemma 9 on the support face in the boundary case, and the singleton convention when $|B_o| = 1$. Therefore

$$F_q(\mu_{q, P_1 \triangleright \{Q^o\}}) = F_q(\mu_{q, P_1}) + \sum_{o: m(o) > 0} m(o) \beta(q, r_o) F_{r_o}(\mu_{r_o, Q^o}),$$

where $\beta(q, r_o)$ denotes the Case 3 coefficient when $B_o = A$, the Case 2 full-to-face coefficient when $1 < |B_o| < |A|$, and any positive singleton convention when $|B_o| = 1$. $\square$

Lemma 18 (Branch-coefficient cocycle and normalised chain rule). *Let the full-support and full-to-face coefficients be those constructed in Lemma 17, applied in the relevant ambient simplex or intrinsically on the relevant support face. If q has full support and r, s are nondegenerate priors with*

$$\text{supp}(s) \subseteq \text{supp}(r) \subseteq \text{supp}(q),$$

then the coefficients satisfy the cocycle identity

$$\beta(q, s) = \beta(q, r) \beta(r, s).$$

Hence $\beta(q, r) = a_q/a_r$ for positive scales a_q , with singleton scales fixed by convention. The rescaled values

$$\hat{F}_q(\mu) := \frac{F_q(\mu)}{a_q}$$

satisfy the exact chain rule: for every first-stage channel P_1 at full-support q and every continuation profile $\{Q^o\}_{o: m(o) > 0}$,

$$\hat{F}_q(\mu_{q, P_1 \triangleright \{Q^o\}}) = \hat{F}_q(\mu_{q, P_1}) + \sum_{o: m(o) > 0} m(o) \hat{F}_{r_o}(\mu_{r_o, Q^o}).$$

Proof. Full-support cocycle. First consider reachable full-support $q, r, s \in \text{int } \Delta(A)$ with $|A| \geq 2$. For such priors, all three tangent spaces coincide. By the tangent-space identification in Lemma 17, $V_q = V_r = V_s = W$ for all full-support priors. The full-support coefficient identities constructed there share a common domain W :

$$L_q|_W = \beta(q, r) L_r|_W, \quad L_r|_W = \beta(r, s) L_s|_W.$$

Substituting the second into the first:

$$L_q|_W = \beta(q, r)\beta(r, s)L_s|_W.$$

Comparing with $L_q|_W = \beta(q, s)L_s|_W$ and using $L_s \neq 0$:

$$\beta(q, s) = \beta(q, r)\beta(r, s).$$

Reachability. The above uses the full-support coefficient construction for the pairs (q, r) , (r, s) , and (q, s) . Each invocation requires the pair to be reachable: the second prior must be a possible posterior of some experiment at the first prior. For any full-support q, r , construct a two-outcome experiment P_1 with $P_1(1 | a) = \varepsilon r(a)/q(a)$ for $0 < \varepsilon < \min\{1, \min_a q(a)/r(a)\}$. Outcome 1 has probability ε and posterior r ; outcome 0 has posterior $(q - \varepsilon r)/(1 - \varepsilon)$. Hence any full-support r is reachable from any full-support q , so the construction applies to all full-support pairs. The cocycle therefore holds for all full-support triples.

Fix a full-support basepoint q_0 , set $a_{q_0} = 1$, and for full-support q define

$$a_q := \frac{1}{\beta(q_0, q)}.$$

By the full-support cocycle, this is equivalently $a_q = \beta(q, q_0)$, and

$$\beta(q, r) = \frac{\beta(q, q_0)}{\beta(r, q_0)} = \frac{a_q}{a_r}.$$

Boundary coefficients and face scales. If r is nondegenerate with support $B \subsetneq A$, Lemma 17 constructs the ambient full-to-face coefficient $\beta_A(q, r) > 0$ by

$$L_q^A i_B = \beta_A(q, r) L_r^B.$$

When the ambient action set is clear, write this coefficient as $\beta(q, r)$.

When nondegenerate boundary priors r, s have common support B , $\beta_A(r, s)$ means the unique $c > 0$ satisfying

$$L_r^A(i_B \sigma) = c L_s^A(i_B \sigma) \quad (\sigma \in W_B).$$

This is a scalar on the common face, not on all of W_A . Since

$$L_r^A \circ i_B = L_{r|_B}^B, \quad L_s^A \circ i_B = L_{s|_B}^B,$$

and the intrinsic face scalar satisfies $L_{r|_B}^B(\sigma) = \beta_B(r|_B, s|_B) L_{s|_B}^B(\sigma)$, we have

$$\beta_A(r, s) = \beta_B(r|_B, s|_B).$$

Fix the same full-support basepoint q_0 on $\Delta(A)$. If r_0 is any nondegenerate face basepoint with support B , then for every nondegenerate r with support B ,

$$\beta_A(q_0, r) = \beta_A(q_0, r_0) \beta_B(r_0|_B, r|_B).$$

Indeed, both sides are the coefficient multiplying $L_{r|_B}^B$ to obtain $L_{q_0}^A \circ i_B$, and $L_{r|_B}^B \neq 0$. Define the ambient boundary scale by

$$a_r := \frac{1}{\beta_A(q_0, r)}.$$

If the face scale is based at r_0 with $a_{r_0} := 1/\beta_A(q_0, r_0)$, then

$$a_{r|_B}^B := \frac{a_{r_0}}{\beta_B(r_0|_B, r|_B)} = \frac{1}{\beta_A(q_0, r_0) \beta_B(r_0|_B, r|_B)} = \frac{1}{\beta_A(q_0, r)} = a_r.$$

Thus the ambient and intrinsic scales agree: treating r as a boundary posterior in $\Delta(A)$ or as a full-support prior on $\Delta(B)$ yields the same normalised representative $\widehat{F}_r$.

Extension to boundary posteriors. If r is nondegenerate with support $B \subsetneq A$, Lemma 17 defines $\beta(q, r) = \beta_A(q, r)$ by

$$L_q^A i_B = \beta_A(q, r) L_{r|B}^B.$$

Using the full-support identity $L_q^A = \beta(q, q_0) L_{q_0}^A$ and restricting to $i_B(W_B)$,

$$L_q^A i_B = \beta(q, q_0) L_{q_0}^A i_B = \beta(q, q_0) \beta_A(q_0, r) L_{r|B}^B.$$

Comparing with the definition of $\beta_A(q, r)$ and using $L_{r|B}^B \neq 0$ gives

$$\beta(q, r) = \beta(q, q_0) \beta(q_0, r) = \frac{a_q}{a_r},$$

where $a_r = 1/\beta(q_0, r)$ by the boundary-scale definition above.

Singleton scale conventions. If $r = \delta_a$ is a singleton prior, set $a_{\delta_a} := a_{q_0} = 1$ and $\widehat{F}_{\delta_a} \equiv 0$. Define $\beta(q, \delta_a) := a_q/a_{\delta_a}$ as a convention. No behavioural branch coefficient is identified at a singleton target, but singleton terms in the aggregation and chain formulas are always zero.

Mixed-support cocycle. Let r, s be nondegenerate with nested supports $C := \text{supp}(s) \subseteq B := \text{supp}(r)$, and write $s|_B$ for s viewed as a boundary prior on the action set B . Interpret

$$\beta(r, s) := \beta_B(r|_B, s|_B),$$

the intrinsic full-to-face coefficient on action set B . Let $i_C^A : W_C \rightarrow W_A$ and $i_C^B : W_C \rightarrow W_B$ be the face inclusions. For every $\sigma \in W_C$,

$$L_q^A i_C^A \sigma = \beta(q, r) L_{r|B}^B i_C^B \sigma = \beta(q, r) \beta(r, s) L_{s|C}^C \sigma.$$

The direct full-to-face definition from Lemma 17 for the pair (q, s) gives

$$L_q^A i_C^A \sigma = \beta(q, s) L_{s|C}^C \sigma.$$

Since $L_{s|C}^C \neq 0$, $\beta(q, s) = \beta(q, r) \beta(r, s)$. If $\text{supp}(s) \not\subseteq \text{supp}(r)$, then $\beta(r, s)$ is not behaviourally defined; singleton targets are also excluded from the behavioural cocycle because their scales are conventions only.

Chain rule. Applying Lemma 17 and substituting $\beta(q, r) = a_q/a_r$ gives

$$F_q(\mu_{q, P_1 \triangleright \{Q^\circ\}}) = F_q(\mu_{q, P_1}) + \sum_{o: m(o) > 0} m(o) \frac{a_q}{a_{r_o}} F_{r_o}(\mu_{r_o, Q^\circ}).$$

Dividing by a_q and writing $\widehat{F}_q(\mu) := F_q(\mu)/a_q$ yields

$$\widehat{F}_q(\mu_{q, P_1 \triangleright \{Q^\circ\}}) = \widehat{F}_q(\mu_{q, P_1}) + \sum_{o: m(o) > 0} m(o) \widehat{F}_{r_o}(\mu_{r_o, Q^\circ}). \quad \square$$

Lemma 19 (Coherent relabelling and face scales). *The scales produced by Lemma 18 can be chosen simultaneously across all finite action sets so that:*

1. for every bijection $\pi : A \rightarrow A'$,

$$a_{q\pi}^{A'} = a_q^A;$$

2. if $\iota : B \hookrightarrow A$ is an injection with $|B| \geq 2$, r has full support on B , and q has full support on A , then the full-to-face branch coefficient for reaching the face $\iota(B)$ is

$$\beta_\iota(q, r) = \frac{a_q^A}{a_r^B}. \quad (\text{FS})$$

Singleton target scales remain a convention because every continuation value on a singleton fibre is zero.

Proof. For every nondegenerate finite action set A , first choose the full-support scales supplied by the cocycle part of Lemma 18, so that

$$\beta_A(q, q') = \frac{a_q^A}{a_{q'}^A} \quad (q, q' \in \text{int } \Delta(A)).$$

They are unique up to multiplication by one positive constant depending on A . Normalise initially by $a_{u_A}^A = 1$, where u_A is uniform. Exact relabelling invariance of the representatives implies relabelling invariance of the uniquely determined branch coefficients. Since bijections carry uniform priors to uniform priors, the initial scales are exactly invariant under bijections.

Let $\iota : B \hookrightarrow A$ be an injection between nondegenerate finite sets. By relabelling B onto its image $\iota(B)$ and using Corollary 15, it is enough to define the coefficient for the inclusion of the face $\iota(B)$; the notation below keeps the original symbol ι . Let i_ι denote the induced map on signed posterior-law tangent spaces. For full-support $q \in \Delta(A)$ and $r \in \Delta(B)$, let $\beta_\iota(q, r) > 0$ be the full-to-face scalar characterised by

$$L_q^A \circ i_\iota = \beta_\iota(q, r) L_r^B \quad \text{on } W_B.$$

The number

$$K(\iota) := \beta_\iota(q, r) \frac{a_r^B}{a_q^A} \tag{K}$$

is independent of both q and r . Independence of q follows by comparing two full-support ambient priors and using $L_q^A = (a_q^A/a_{q'}^A)L_{q'}^A$ on W_A . Independence of r follows from same-face compatibility: if r, s have full support on B , then $L_r^B = (a_r^B/a_s^B)L_s^B$, and hence $\beta_\iota(q, s) = \beta_\iota(q, r)a_r^B/a_s^B$.

The defects multiply under composition. If $C \xhookrightarrow{j} B \xhookrightarrow{\iota} A$ and all three sets in the comparison are nondegenerate, then restricting first to B and then to C gives, for full-support q, r, s on A, B, C respectively,

$$\beta_{\iota \circ j}(q, s) = \beta_\iota(q, r)\beta_j(r, s).$$

Substituting the definition of K and cancelling the intermediate scale a_r^B gives

$$K(\iota \circ j) = K(\iota)K(j). \tag{KC}$$

Exact relabelling invariance shows that $K(\iota)$ depends only on $n := |A|$ and $m := |B|$; write it as $K_{n,m}$. Bijections give $K_{n,n} = 1$, and

$$K_{n,\ell} = K_{n,m}K_{m,\ell} \quad (2 \leq \ell \leq m \leq n).$$

Put $t_n := K_{n,2}$, with $t_2 = 1$. Then $K_{n,m} = t_n/t_m$. Finally replace every scale on an n -point action set by

$$\tilde{a}_q^A := t_n a_q^A.$$

Under this change, the defect in (K) becomes

$$\widetilde{K}_{n,m} = K_{n,m} \frac{t_m}{t_n} = 1.$$

The rescaling is common to all priors on a fixed action set, so it leaves all within-set cocycle ratios unchanged; because t_n depends only on cardinality, it also preserves exact relabelling invariance. Dropping tildes gives (FS). Singleton target scales may be set arbitrarily because all singleton continuation terms are zero. $\square$

Lemma 20 (Interaction collapse and universal chain scale). *The universal interaction coefficient in (QA) is zero. Moreover, there exists $a > 0$ such that the coherently chosen chain scales satisfy $a_q = a$ for every nondegenerate full-support prior on every finite action set. The same value may be assigned to singleton priors. Consequently,*

$$F_{p \otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) \quad (\text{PA})$$

and the branch formula has a universal chain scale.

Proof. Define, for every finite prior q read on its support,

$$H(q) := F_q(\chi_q), \quad Z(q) := 1 + \kappa H(q).$$

For a nondegenerate prior, local non-triviality gives $H(q) > 0$, and (POS) gives

$$Z(q) > 0. \quad (\text{Z+})$$

For a singleton, $H(q) = 0$ and $Z(q) = 1$.

Step 1: product revelation links the chain scale to Z . Let $q \in \Delta(A)$ and $r \in \Delta(B)$ be nondegenerate and full support. Quasi-additivity applied to full revelation gives

$$H(q \otimes r) = H(q) + H(r) + \kappa H(q)H(r). \quad (\text{QH})$$

Compute the same value sequentially. First reveal the A -coordinate and then reveal the B -coordinate in every branch. The first-stage value is $F_{q \otimes r}(\chi_q \otimes \delta_r) = H(q)$ by (QA). After observing a , the posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify its continuation full-revelation value with $H(r)$. For the injection

$$\iota_a : B \hookrightarrow A \times B, \quad b \mapsto (a, b),$$

Lemma 19 gives the branch coefficient

$$\beta_{\iota_a}(q \otimes r, r) = \frac{a_{q \otimes r}^{A \times B}}{a_r^B}.$$

The branch weights are $q(a)$ and the same coefficient and continuation value appear in every branch, so $\sum_a q(a) = 1$. Lemma 17 therefore yields

$$H(q \otimes r) = H(q) + \frac{a_{q \otimes r}}{a_r} H(r). \quad (\text{SA})$$

Comparing (QH) and (SA), and using $H(r) > 0$,

$$\frac{a_{q \otimes r}}{a_r} = Z(q). \quad (\text{R1})$$

Revealing B first gives symmetrically

$$\frac{a_{q \otimes r}}{a_q} = Z(r). \quad (\text{R2})$$

Hence $a_r Z(q) = a_q Z(r)$, so there is a universal constant $C > 0$ such that

$$a_q = CZ(q) \quad (\text{AS})$$

for every nondegenerate full-support prior on every finite action set. Assign $a_q := C$ at singleton priors, where $Z(q) = 1$ and all continuation values are zero.

If $\kappa = 0$, then $Z \equiv 1$, equation (AS) already gives universal scales, and (QA) is exact product additivity. For the remaining argument suppose $\kappa \neq 0$.

Step 2: the induced weight equation. Let q be full support on A , let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition into nonempty cells, and put

$$p_i := q(A_i), \quad q^i := q(\cdot \mid A_i), \quad p := (p_1, \dots, p_m).$$

Let $G_{\mathcal{P}}$ reveal the block. Undetectable-distinctions neutrality gives $(q, G_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}})$. The early cross-prior representation Lemma 16, importantly for the *unscaled* F 's, therefore gives

$$F_q(\mu_{q, G_{\mathcal{P}}}) = H(p). \quad (\text{BG})$$

Append full revelation in every branch. The compound posterior law is χ_q , so the chain formula and (AS) give

$$H(q) = H(p) + Z(q) \sum_{i=1}^m p_i \frac{H(q^i)}{Z(q^i)}. \quad (\text{GR})$$

Define

$$w(q) := \frac{1}{Z(q)} > 0.$$

Using $\kappa H(x) = Z(x) - 1$ in (GR) gives

$$\frac{Z(p)}{Z(q)} = \sum_{i=1}^m p_i \frac{1}{Z(q^i)}.$$

Equivalently,

$$w(q) = w(p) \sum_{i=1}^m p_i w(q^i). \quad (\text{W})$$

No continuity assertion is needed here.

Step 3: a two-grouping argument forces $w \equiv 1$. All distributions in this paragraph are read on their positive-probability supports, so the weight equation applies after support restriction. First, (W) applied to a product distribution gives

$$w(u \otimes v) = w(u)w(v), \quad (\text{M})$$

because the first-coordinate block probabilities are u and every conditional distribution is v .

Take arbitrary finite probability vectors u and v , possibly on different alphabets, and write

$$x := w(u), \quad y := w(v), \quad p := (1/2, 1/2).$$

On the labelled disjoint union $(U \times U)^0 \sqcup (V \times V)^1$, form the distribution

$$T := \frac{1}{2}(u \otimes u)^0 \sqcup \frac{1}{2}(v \otimes v)^1.$$

Grouping first by the top block and using (M),

$$w(T) = w(p) \frac{x^2 + y^2}{2}. \quad (\text{E1})$$

Now group first by the pair consisting of the top-block label and the first within-block coordinate. Its marginal is

$$S := \frac{1}{2}u^0 \sqcup \frac{1}{2}v^1,$$

and (W) gives

$$w(S) = w(p) \frac{x + y}{2}.$$

Conditional on a point in the first part of S , the remaining coordinate has distribution u ; conditional on a point in the second part, it has distribution v . A second application of (W) therefore gives

$$w(T) = w(S) \frac{x+y}{2} = w(p) \left(\frac{x+y}{2} \right)^2. \quad (\text{E2})$$

Since $w(p) > 0$, comparison of (E1) and (E2) yields

$$\frac{x^2 + y^2}{2} = \left(\frac{x+y}{2} \right)^2,$$

so $(x-y)^2 = 0$. As u and v were arbitrary, w is constant on all finite probability vectors. At a singleton prior, $H = 0$, hence $w = 1$. Therefore $w(q) \equiv 1$.

Thus $Z(q) = 1$ and $\kappa H(q) = 0$ for every q . Local non-triviality gives a nondegenerate q with $H(q) > 0$, hence $\kappa = 0$, contradicting the temporary alternative unless the interaction coefficient is zero. Equation (AS) now gives $a_q = C$ universally. Finally (QA) reduces to exact product additivity (PA). $\square$

Lemma 21 (Entropy reduction and Faddeev). *By Lemma 20, $a_q \equiv a$ is universal. Define*

$$\hat{F}_q := F_q/a, \quad H(q) := \hat{F}_q(\chi_q),$$

the rescaled value of full revelation. Then, for every channel P ,

$$\hat{F}_q(\mu_{q,P}) = H(q) - \sum_{o: m_{q,P}(o) > 0} m_{q,P}(o) H(r_o^{q,P}). \quad (\text{ER})$$

Moreover H satisfies Faddeev's recursion, so

$$H(q) = \alpha H_{\text{Sh}}(q)$$

for some $\alpha > 0$. Consequently,

$$\hat{F}_q(\mu_{q,P}) = \alpha I_{q,P}(A; O).$$

Proof. Entropy-reduction formula. Boundary values of H are read through support restriction: if q has support B , then $H(q) := H(q|_B)$, and $H(\delta_a) = 0$. Let $P : A \rightarrow \Delta(O)$ be any channel at full-support prior q . Append full revelation $\text{Id}_A : A \rightarrow \Delta(A)$ after P : the compound $P \triangleright \{\text{Id}_A\}_o$ (using Id_A in every branch) has $\mu_{r_o, \text{Id}_A} = \chi_{r_o}$ and the overall posterior law is full revelation χ_q . Applying the chain rule (Lemma 18) gives

$$\hat{F}_q(\chi_q) = \hat{F}_q(\mu_{q,P}) + \sum_{o: m(o) > 0} m(o) \hat{F}_{r_o}(\chi_{r_o}).$$

Hence

$$\hat{F}_q(\mu_{q,P}) = H(q) - \sum_{o: m(o) > 0} m(o) H(r_o).$$

If q is on the boundary, restrict q and P to $B = \text{supp}(q)$, apply the full-support formula on $\Delta(B)$, and pull the value and comparison back by Lemma 12; zero-probability rows and outcomes do not affect either side of the displayed identity.

Scaled cross-prior bridge. Lemma 16 was proved for the unscaled values F_q . Since Lemma 20 has made $a_q \equiv a$ universal, it is equivalently valid for

$$V(q, P) := \hat{F}_q(\mu_{q,P}) = F_q(\mu_{q,P})/a :$$

for all finite priors and channels,

$$q^0 \succeq_{P \sqcup Q} r^1 \iff V(q, P) \geq V(r, Q).$$

Faddeev recursion. First let q be full support and let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition into nonempty cells. Then $p_i := q(A_i) > 0$ for every i . Let $G_{\mathcal{P}} : A \rightarrow \Delta(\{1, \dots, m\})$ reveal only the block index: $G_{\mathcal{P}}(i | a) = \mathbf{1}_{a \in A_i}$. Write

$$p_i := q(A_i), \quad q^i := q(\cdot | A_i),$$

and let $p = (p_1, \dots, p_m) \in \Delta(\{1, \dots, m\})$. Consider the deterministic kernel $S : A \rightarrow \Delta(\{1, \dots, m\})$ given by $S(i | a) = \mathbf{1}_{a \in A_i}$, so $qS = p$.

By Lemma 11, since $G_{\mathcal{P}}$ is S -measurable,

$$(q, G_{\mathcal{P}}) \sim (p, \text{Id}_{\{1, \dots, m\}})$$

in the two-block comparison. Applying the scaled form of Lemma 16 to the forward comparison gives $V(q, G_{\mathcal{P}}) \geq V(p, \text{Id}_{\{1, \dots, m\}})$. Applying the reverse-block form from Lemma 3 and then the same bridge to the reverse comparison gives $V(p, \text{Id}_{\{1, \dots, m\}}) \geq V(q, G_{\mathcal{P}})$. Therefore

$$\hat{F}_q(\mu_{q, G_{\mathcal{P}}}) = \hat{F}_p(\chi_p) = H(p).$$

The entropy-reduction formula applied to $G_{\mathcal{P}}$ gives

$$H(q) = \hat{F}_q(\chi_q) = \hat{F}_q(\mu_{q, G_{\mathcal{P}}}) + \sum_{i=1}^m p_i \hat{F}_{q^i}(\chi_{q^i}) = H(p) + \sum_{i=1}^m p_i H(q^i).$$

This is Faddeev's recursion. For boundary q , restrict to $\text{supp}(q)$, apply the same full-support recursion to the nonempty positive-mass intersections, and pull back by Lemma 12; equivalently, the recursion is read over positive-mass blocks only.

We now verify that H is continuous on each $\Delta_n := \{q \in \Delta(A) : |A| = n\}$.

Binary continuity via erasure calibrators. Define the binary entropy function $h : [0, 1] \rightarrow \mathbb{R}$ by $h(t) := H(t, 1 - t)$. We show h is continuous using prior-independent calibrators. By Lemma 9, support restriction, and $a > 0$, $H(q) \geq 0$ for every q , so $h(t) \geq 0$ on $[0, 1]$. If $c < 0$, then

$$\{t : h(t) \geq c\} = [0, 1], \quad \{t : h(t) \leq c\} = \emptyset,$$

both closed. Now fix $c \geq 0$. By local non-triviality, there exists a binary prior $\theta = (s, 1 - s)$ with $u := H(\theta) > 0$. Choose $n \in \mathbb{N}$ and $\lambda \in [0, 1]$ such that $c = \lambda nu$ (take n large enough that $c \leq nu$, then set $\lambda := c/(nu)$).

Let $w_n := \theta^{\otimes n}$ be the n -fold product prior on $B_n := \{0, 1\}^n$. By product additivity, $H(w_n) = nH(\theta) = nu$. Define the erasure experiment $E_{n, \lambda} : B_n \rightarrow \Delta(\{*, 1, \dots, 2^n\})$ which reveals the state fully with probability λ and reveals nothing with probability $1 - \lambda$. By the entropy-reduction formula,

$$\hat{F}_{w_n}(\mu_{w_n, E_{n, \lambda}}) = \lambda H(w_n) = \lambda nu = c.$$

Now consider the block action space $A_c := \{0, 1\} \sqcup B_n$ and the block-diagonal channel $K_c := \text{Id}_{\{0, 1\}} \sqcup E_{n, \lambda}$. For $t \in [0, 1]$, let q_t^0 be the prior $(t, 1 - t)$ supported on block $\{0, 1\}$, and let w_n^1 be w_n supported on block B_n . By the scaled form of Lemma 16 (which extends to boundary priors via Lemma 12),

$$q_t^0 \succeq_{K_c} w_n^1 \iff V(q_t, \text{Id}_{\{0, 1\}}) = h(t) \geq c = V(w_n, E_{n, \lambda}).$$

Since K_c is fixed and the map $t \mapsto q_t^0$ is continuous in ℓ_1 , Axiom (A2) implies that the set $\{t \in [0, 1] : h(t) \geq c\}$ is closed. Similarly, $\{t \in [0, 1] : h(t) \leq c\}$ is closed. Thus all real superlevel and sublevel sets of h are closed, so h is both upper and lower semicontinuous on $[0, 1]$, hence continuous.

Continuity on higher simplices by induction. For $n \geq 3$, write any $q \in \Delta_n$ as $q = (q_1, (1 - q_1)\bar{q})$ where $\bar{q} := (q_2/(1 - q_1), \dots, q_n/(1 - q_1)) \in \Delta_{n-1}$ for $q_1 < 1$. The Faddeev recursion with the binary partition $\{\{1\}, \{2, \dots, n\}\}$ gives

$$H(q) = h(q_1) + (1 - q_1)H(\bar{q}).$$

On the region $0 < q_1 < 1$, this is continuous: h is continuous by binary continuity, $H|_{\Delta_{n-1}}$ is continuous by the induction hypothesis, and $q \mapsto (q_1, \bar{q})$ is continuous. If $q_1 = 0$, support restriction, equivalently the positive-mass-block convention in the recursion, gives $H(q) = H(\bar{q})$ and $h(0) = 0$, so continuity at that face follows from the induction hypothesis. At $q_1 = 1$, the induction hypothesis and compactness of Δ_{n-1} imply H is bounded on Δ_{n-1} , so $(1 - q_1)H(\bar{q}) \rightarrow 0$ as $q_1 \rightarrow 1$. Thus $H(q) \rightarrow h(1) = 0 = H(\delta_1)$. The other boundary faces are handled by support restriction, relabelling on the support, and pulling back. This completes the induction.

We have verified the hypotheses of the strong-additivity form of Faddeev's finite entropy characterisation, as stated for example by Baez–Fritz–Leinster [6, Theorem 6], following Faddeev's original theorem [19]: a function on finite probability vectors that is continuous on each closed simplex, invariant under bijections, expansive under adjoining zero-probability states, zero on point masses, and strongly additive is a nonnegative multiple of Shannon entropy. Here invariance under bijections follows from Corollary 15, with boundary cases obtained by restricting to the support and pulling back; expansibility follows from support restriction; and the grouping recursion above is precisely strong additivity, read over positive-mass blocks. Faddeev's theorem therefore gives $H(q) = \alpha H_{\text{Sh}}(q)$ for some $\alpha \geq 0$. Local non-triviality (Axiom (A1)) forces $\alpha > 0$.

Mutual information. Substituting $H = \alpha H_{\text{Sh}}$ into the entropy-reduction formula yields

$$\hat{F}_q(\mu_{q,P}) = \alpha H_{\text{Sh}}(q) - \sum_{o: m_{q,P}(o) > 0} m_{q,P}(o) \alpha H_{\text{Sh}}(r_o^{q,P}) = \alpha (H_{\text{Sh}}(q) - \mathbb{E}[H_{\text{Sh}}(r_O)]) = \alpha I_{q,P}(A; O).$$

□

Lemma 22 (Fixed-environment representation). *For every finite channel $P : A \rightarrow \Delta(O)$ and all $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{q,P}(A; O) \geq I_{q',P}(A; O).$$

Proof. Let

$$V(s, R) := \hat{F}_s(\mu_{s,R}),$$

with boundary priors interpreted by Lemma 12. Lemma 16, divided by the universal scale from Lemma 20, gives the cross-prior block-comparison bridge: for finite action sets, priors s, t , and channels R, S ,

$$s^0 \succeq_{R \sqcup S} t^1 \iff V(s, R) \geq V(t, S),$$

including boundary priors by support restriction. Lemma 21 gives, for every finite prior/channel pair,

$$V(s, R) = \alpha I_{s,R}$$

with $\alpha > 0$.

Fix $P : A \rightarrow \Delta(O)$ and $q, q' \in \Delta(A)$. By the first clause of Axiom (A3),

$$q \succeq_P q' \iff q^0 \succeq_{P \sqcup P} (q')^1.$$

Applying this bridge with $(s, R) = (q, P)$ and $(t, S) = (q', P)$ gives

$$q^0 \succeq_{P \sqcup P} (q')^1 \iff V(q, P) \geq V(q', P).$$

Substituting $V = \alpha I$ and using $\alpha > 0$ yields

$$q \succeq_P q' \iff I_{q,P}(A; O) \geq I_{q',P}(A; O).$$

If q or q' is on the boundary, the bridge is read after restricting each pair to its positive-probability support and then pulling the comparison back by Lemma 12. Deleting zero-probability actions does not change mutual information, and the Axiom (A3) equivalence remains the ambient fixed-channel comparison for P . $\square$

Worked example

Example 23 (Donor attribution). A donor can route a gift either to a named project N or to a pooled fund F :

$$A = \{N, F\}.$$

The public record is binary. It either contains a project-linked report L or only an aggregate report U :

$$O = \{L, U\}.$$

Let

$$P(L | N) = \alpha, \quad P(L | F) = \beta, \quad 0 \leq \beta \leq \alpha \leq 1.$$

When $\beta < \alpha$, the named project is more likely to leave a project-linked trace, but pooled funding may also occasionally be linked and named projects may sometimes disappear into aggregate reporting. Write $q(N) = p$ and $q(F) = 1 - p$. The probability of a linked report is

$$m_{q,P}(L) = \beta + (\alpha - \beta)p.$$

Therefore

$$I_{q,P}(A; O) = h(\beta + (\alpha - \beta)p) - p h(\alpha) - (1 - p)h(\beta),$$

where

$$h(x) := -x \log x - (1 - x) \log(1 - x)$$

is binary entropy, with the convention $0 \log 0 = 0$.

The formula isolates attribution rather than payoff. If $\alpha = \beta$, the public record is independent of the donor's act and traceability is zero for every q . If $\alpha = 1$ and $\beta = 0$, the record fully reveals the realised act and traceability is $h(p)$. If $p = 0$ or $p = 1$, traceability is again zero, because there is no uncertainty about the act for the report to resolve. Thus traceability comes from statistical separation in the reporting distributions together with uncertainty about which act was realised.

Discussion

Three points are worth emphasising.

First, the theorem is *within*-environment in its behavioural content. The decision-maker chooses q while the channel P indexes the decision problem. Cross-problem-looking comparisons are routed through block comparison environments, so they remain ordinary comparisons between lotteries in one fixed environment. This is analogous to the role played by statewise coherence principles in classic decision theory: they constrain how one and the same behavioural criterion is represented across transformed problems without turning the environment into an object of choice.

Second, the alignment of the fibrewise cardinal scales, which precedes the entropy/Faddeev identification, is driven by branchwise continuation monotonicity together with the data-processing axioms, and not by any separate axiom on product environments. Background inertness (Lemma 13) identifies the coordinate slice orders on product posterior domains. Separate affinity then yields the quasi-additive product form in Lemma 14; branch aggregation,

face-scale coherence, and the two-grouping argument in Lemma 20 eliminate the possible interaction term.

Third, the representation is not an expected-utility theorem. If two mechanisms induce the same lottery on final payoffs but different joint laws of action and outcome, expected utility is indifferent while the present model need not be. In that sense the model is non-consequentialist over payoff outcomes alone. This fixed-environment formulation places the paper close in spirit to the behavioural literatures on preference for flexibility [31], temptation and self-control [26], and the intrinsic value of decision rights [5], while the mathematical backbone remains the information-theoretic and statistical-experiment traditions of Blackwell [10] and Faddeev [19]. One can, however, combine the traceability term with ordinary utility over final outcomes:

$$U_P(q) = \mathbb{E}_{q,P}[u(O)] + \lambda I_{q,P}(A; O).$$

The fixed-environment theorem identifies the second term.

Sign duality: traceability and deniability. A final observation concerns the role of the signed axioms. The axiom system splits into a structural part and an oriented part. Axioms (A2) and (A3) are equivalences or closedness conditions and are invariant under reversal of every preference in the family; Axiom (A6) is self-dual, since branchwise monotonicity for the reversed family is the original axiom with the two continuation profiles exchanged, and strictness survives the exchange. Only the orientation clause of Axiom (A1) and the two data-processing axioms (A4) and (A5) carry a sign. Consequently, a family $\{\succeq_P\}_P$ satisfies the structural axioms together with the reversed forms of (A1), (A4), and (A5) if and only if its reversed family $\{\preceq_P\}_P$ satisfies Axioms (A1)–(A6), which by Theorem 2 holds if and only if

$$q \succeq_P q' \iff I_{q,P}(A; O) \leq I_{q',P}(A; O).$$

The reversed system is therefore represented by negative mutual information. The two orientations cannot be mixed: reversed (A4)–(A5) together with the original orientation of (A1) force the uninformative channel weakly above full revelation, contradicting the strict ranking, and symmetrically for the opposite mixture. Hence the structural axioms admit exactly three coherent completions: total indifference, mutual information, and its negative.

The negative orientation has a natural interpretation: a preference for *deniability*, under which the realised act should leave as little statistical trace in outcomes as possible. Two qualifications matter for that reading. First, a degenerate lottery achieves the deniability optimum trivially, since a known act cannot be revealed; the criterion values the outcome teaching observers nothing *new*, not the act being unknown as such. Second, the object concealed is the realised act in a one-shot environment, not a persistent characteristic of the agent, so the criterion is closer to untraceability than to privacy in the ordinary sense.

The dichotomy also clarifies what is behavioural in the present system. In the instrumental theory of information, Blackwell monotonicity is a theorem: a decision-maker who can condition on a signal is weakly better off with a finer signal because refinements can be ignored. Here the corresponding conditions, Axioms (A4) and (A5), are primitive postulates about taste, and the coherence of their reversal shows that they are substantive rather than formal: the structural axioms force any coherent non-instrumental attitude towards the legibility of one’s own acts onto the mutual-information scale, and the sign of the oriented axioms selects between traceable agency and deniability.

Relation to empowerment. A remark connects the representation to the empowerment literature in artificial intelligence and complex systems, where an agent’s empowerment is defined as the Shannon capacity of the channel from its actions to its future observations, $C(P) = \max_q I_{q,P}(A; O)$, and is interpreted as a universal measure of control [30, 38]; related

free-energy approaches likewise take variational information quantities as primitive objectives [22]. An agent with the preferences characterised here, left free to choose any lottery in $\Delta(A)$, would select the capacity-achieving input distribution: by Theorem 2, $\max_q I_{q,P}(A; O)$ is exactly her most preferred behaviour in environment P , so empowerment is the indirect-utility value of traceable agency. The relationship also marks the difference. Empowerment is a property of the environment—the capacity of its act-to-outcome channel—taken as a primitive objective, and its maximisation is interpreted as a drive for control. The present theorem is about the preferences underneath: it represents the whole map $q \mapsto I_{q,P}(A; O)$, so the value of every behaviour is determined, not only the optimum, and constrained or payoff-tempered behaviour, as in the combination $\mathbb{E}_{q,P}[u(O)] + \lambda I_{q,P}(A; O)$ above, remains well defined. Moreover, the interpretation forced by the axioms is informational rather than causal: what the axioms discipline is the statistical trace of the act in the outcome, which coincides with influence only when correlation and control coincide. In this sense the empowerment objective is the optimised shadow of a preference for traceable agency, and the present result supplies the behavioural foundation that the empowerment literature posits directly.

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